

LAPORAN PRAKTIKUM 1

SISTEM OPERASI



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BAB I

PENDAHULUAN

LATAR BELAKANG

Sistem operasi merupakan perangkat lunak inti yang mengatur dan mengendalikan seluruh sumber daya komputer, baik perangkat keras maupun perangkat lunak. Tanpa sistem operasi, pengguna tidak dapat berinteraksi dengan komputer secara efektif karena semua proses seperti pengelolaan memori, CPU, penyimpanan, dan perangkat I/O dijalankan serta dikendalikan oleh sistem operasi. Menurut Silberschatz (2020), sistem operasi berperan sebagai pengelola sumber daya (*resource manager*) dan penyedia antarmuka antara pengguna serta perangkat keras.

Melalui praktikum Sistem Operasi ini, mahasiswa mempelajari secara langsung bagaimana sistem operasi bekerja, khususnya dalam lingkungan Linux. Materi yang dipelajari meliputi layanan sistem operasi (*Operating-System Services*), antarmuka pengguna dan sistem (*User and Operating-System Interface*), serta pemanggilan sistem (*System Calls*) yang menjadi sarana komunikasi antara program dan kernel. Selain itu, mahasiswa juga diperkenalkan pada *System Services*, proses *linking* dan *loading*, serta struktur dan desain sistem operasi modern.

Dalam praktikum ini, mahasiswa menerapkan konsep-konsep tersebut melalui pembuatan program sederhana menggunakan bahasa C untuk memahami hubungan antara shell, kernel, dan aplikasi. Melalui kegiatan ini, mahasiswa memperoleh pemahaman yang lebih mendalam tentang cara kerja sistem operasi dalam mengelola sumber daya, menjalankan proses, serta menjaga stabilitas sistem. Dengan demikian, praktikum ini memberikan dasar penting bagi mahasiswa Teknik Informatika dalam memahami mekanisme kerja sistem komputer secara menyeluruh dan efisien.

RUMUSAN MASALAH

1. Bagaimana sistem operasi bekerja dalam mengelola sumber daya komputer seperti proses, memori, dan file?
2. Bagaimana mekanisme antarmuka antara pengguna, shell, dan kernel melalui Command Line Interface (CLI) serta system calls?
3. Bagaimana penerapan konsep layanan sistem operasi dapat dipahami melalui praktik langsung menggunakan program C pada lingkungan Linux?

TUJUAN

1. Untuk memahami fungsi dan cara kerja sistem operasi dalam mengatur serta mengoordinasikan sumber daya komputer.
2. Untuk mempelajari proses komunikasi antara pengguna, shell, dan kernel melalui perintah Linux dan implementasi system calls.
3. Untuk meningkatkan kemampuan praktis mahasiswa dalam mengaplikasikan teori sistem operasi melalui eksperimen dan pemrograman sederhana di Linux.

BAB II

PEMBAHASAN

2.1 Operating-System Services

Layanan sistem operasi (Operating-System Services) adalah fungsi penting yang disediakan OS untuk mendukung program dan pengguna. Praktik ini menunjukkan cara OS bekerja melalui perintah Linux dan program C yang menampilkan layanan seperti eksekusi proses, operasi I/O, manipulasi file, komunikasi, deteksi kesalahan, serta alokasi sumber daya secara efisien.

2.1.1 Menjelajahi Layanan OS

- Lihat informasi system

```
asus@LAPTOP-BV040HF1:~$ uname -a
Linux LAPTOP-BV040HF1 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
asus@LAPTOP-BV040HF1:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:        24.04
Codename:       noble
```

- Lihat proses yang berjalan

```
asus@LAPTOP-BV040HF1:~$ ps aux | head -20
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1 21688 11988 ?        Ss   20:51   0:00 /sbin/init
root         2  0.0  0.0   3872  1664 ?        Sl   20:51   0:00 /init
root         7  0.0  0.0   3872  1792 ?        Sl   20:51   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root        41  0.2  0.1  42164 15744 ?        S<s  20:51   0:00 /usr/lib/systemd/systemd-journald
root        91  0.2  0.0  25892  6272 ?        Ss   20:51   0:00 /usr/lib/systemd/systemd-udevd
systemd+  112  0.1  0.1  21456 12544 ?        Ss   20:51   0:00 /usr/lib/systemd/systemd-resolved
systemd+  116  0.1  0.0  91824  7424 ?        Ssl  20:51   0:00 /usr/lib/systemd/systemd-timesyncd
root       177  0.0  0.0   4236  2560 ?        Ss   20:51   0:00 /usr/sbin/cron -f -P
message+  178  0.0  0.0   9592  4736 ?        Ss   20:51   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
og-only
root       185  0.1  0.1  17960  8192 ?        Ss   20:51   0:00 /usr/lib/systemd/systemd-logind
root       187  0.1  0.1 1756096 12832 ?        Ssl  20:51   0:00 /usr/libexec/mtl-pro-service -vv
root       197  0.0  0.0   3160  1920 hvc0    Ss+  20:51   0:00 /sbin/agetty -o -p -- \u --noclear - 115200,38400,9600 vt220
root       210  0.0  0.0   3116  1792 ttyl    Ss+  20:51   0:00 /sbin/agetty -o -p -- \u --noclear - linux
syslog    212  0.1  0.0 2225088 5504 ?        Ssl  20:51   0:00 /usr/sbin/rsyslogd -n -iNONE
root       218  0.1  0.2 107032 22656 ?        Ssl  20:51   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
gnal
root       307  0.0  0.0   3076  1024 ?        Ss   20:51   0:00 /init
root       308  0.0  0.0   3092  1028 ?        S   20:51   0:00 /init
asus      309  0.0  0.0   6872  5248 pts/0    Ss   20:51   0:00 -bash
root      310  0.0  0.0   6820  4096 pts/1    Ss   20:51   0:00 /bin/login -f
```

- Lihat penggunaan memori

```
asus@LAPTOP-BV040HF1:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.7Gi          497Mi          6.9Gi          3.5Mi          407Mi          7.2Gi
Swap:          2.0Gi           0B           2.0Gi
```

- Lihat informasi CPU

```

asus@LAPTOP-BV84QHF1:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 48 bits physical, 48 bits virtual
Byte order: Little Endian
CPU(s): 16
On-line CPU(s) list: 0-15
Vendor ID: AuthenticAMD
Model name: AMD Ryzen 7 7435HS
CPU family: 25
Model: 68
Thread(s) per core: 2
Core(s) per socket: 8
Socket(s): 1
Stepping: 1
BogoMIPS: 6188.19
Flags: fpu_eme de pae tsc tscn pae mce rdtscp sep mtrr pge mca cmov pat pni36 clflush mwaitx rcp sse sse2 ht syscall nx sseext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good nopl tsc_reliable no_nstap_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 movbe po
pcnt aes xsave avx f16c rdrand hypervisor lahf_lm cap_lgpcy svm_crl_lgpcy sha_ssah aicallipa
sse3dnowprefetch osvw topoext perfctr_core srbdi sbnd ibpb stibp vmcall fsgsbase bmi1 avx2 sm
ap mail error_tscpid pstate mds smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsavei clz
ero xsaveerptr arat npt nrp.save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter p
fthreshold v_vesave_vload umip vaes vpclmulqdq rdpid tsrm

Virtualization features:
Virtualization: AMD-V
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 256 KiB (8 instances)
L1i: 256 KiB (8 instances)
L2: 4 MiB (1 instance)
L3: 16 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node CPU(s): 0-15
Vulnerabilities:
Gather data sampling: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Reg file data sampling: Not affected
Retbleed: Not affected
Spec rstack overflow: Vulnerable: Safe RET, no microcode
Spec store bypass: Mitigation: Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation: usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation: Retpolines; IBPB conditional; IBRS_FW; STIBP always-on; RSB filling; PMSB-EIBRS; No
ot affected; BHI Not affected
Srbds: Not affected
Tsx async abort: Not affected

```

- Lihat perangkat yang terpasang

```

asus@LAPTOP-BV84QHF1:~$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0 388.4M  1 disk
sdb      8:16   0   160M  1 disk
sdc      8:32   0    2G  0 disk [SWAP]
sdd      8:48   0    1T  0 disk /mnt/wslg/distro

```

2.1.2 Program C - Menggunakan Layanan OS

```

asus@LAPTOP-BV84QHF1:~$ nano os_services.c
asus@LAPTOP-BV84QHF1:~$ gcc -o os_services os_services.c
asus@LAPTOP-BV84QHF1:~$ ./os_services
=== Program Execution Service ===
Process ID (PID): 491
Parent Process ID (PPID): 389

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 bytes
File permissions: 644

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully

```

2.2 User and Operating-System Interface

User and Operating-System Interface adalah mekanisme yang memungkinkan pengguna berinteraksi dengan sistem operasi. Melalui Command Line Interface (CLI), pengguna dapat menjalankan perintah seperti navigasi direktori, penggunaan pipelining, redirection, serta menjalankan proses di latar belakang. Pada praktik simple_shell.c, dibuat shell sederhana yang menerima input pengguna, memecah argumen, dan mengeksekusi perintah menggunakan fungsi fork() dan execvp(). Praktik ini memperlihatkan bagaimana antarmuka antara pengguna, shell, dan kernel bekerja untuk menjalankan instruksi secara langsung dan efisien dalam sistem operasi Linux.

2.2.1 Command Line Interface (CLI)

- Navigasi dasar

```
asus@LAPTOP-BV04QHF1:~$ ls -la
total 1064
drwxr-xr-x 7 asus asus 4096 Oct 13 21:17 .
drwxr-xr-x 3 root root 4096 Oct 7 21:22 ..
-rw-r--r-- 1 asus asus 6571 Oct 13 19:30 .bash_history
-rw-r--r-- 1 asus asus 220 Oct 7 21:22 .bash_logout
-rw-r--r-- 1 asus asus 3771 Oct 7 21:22 .bashrc
drwxr-xr-x 2 asus asus 4096 Oct 7 21:24 .cache
drwxr-xr-x 3 asus asus 4096 Oct 10 17:47 .config
-rw-r--r-- 1 asus asus 20 Oct 10 21:06 .lesshst
drwxr-xr-x 3 asus asus 4096 Oct 7 21:51 .local
-rw-rw-r-- 1 asus asus 0 Oct 13 18:33 .motd.shown
-rw-rw-r-- 1 asus asus 807 Oct 7 21:22 .profile
-rw-rw-r-- 1 asus asus 0 Oct 7 22:11 .sudo_as_admin_successful
-rw-rw-r-- 1 root root 178 Oct 9 19:35 'Description=Hy'
-rwxr-xr-x 1 asus asus 17400 Oct 12 19:43 file_monitor
-rw-rw-r-- 1 asus asus 7174 Oct 12 19:42 file_monitor.c
-rw-rw-r-- 1 asus asus 448 Oct 13 18:33 hello_module.c
-rwxr-xr-x 1 asus asus 15144 Oct 10 16:47 libmathops.so
-rw-rw-r-- 1 asus asus 154 Oct 10 12:05 main.c
-rw-rw-r-- 1 asus asus 21452 Oct 10 12:05 main.i
-rw-rw-r-- 1 asus asus 1728 Oct 10 12:08 main.o
-rw-rw-r-- 1 asus asus 910 Oct 10 12:07 main.s
-rw-rw-r-- 1 root root 166 Oct 10 17:11 makefile
-rw-rw-r-- 1 asus asus 118 Oct 10 12:03 math_ops.c
-rw-rw-r-- 1 asus asus 98 Oct 10 11:59 math_ops.h
-rw-rw-r-- 1 asus asus 322 Oct 10 12:05 math_ops.i
-rw-rw-r-- 1 asus asus 1352 Oct 10 16:47 math_ops.o
-rw-rw-r-- 1 asus asus 902 Oct 10 12:07 math_ops.s
-rw-rw-r-- 1 asus asus 785 Oct 9 19:39 my_daemon.c
-rwxr-xr-x 1 asus asus 16488 Oct 13 21:06 os_services
-rw-rw-r-- 1 asus asus 1449 Oct 13 21:06 os_services.c
-rw-rw-r-- 1 asus asus 53 Oct 7 21:59 os_services.c:
-rw-rw-r-- 1 asus asus 761 Oct 10 16:59 os_specific.c
drwxr-xr-x 5 asus asus 4096 Oct 12 19:55 praktikum_os
-rwxr-xr-x 1 asus asus 17040 Oct 11 15:40 process_manager
-rw-rw-r-- 1 asus asus 3785 Oct 11 15:40 process_manager.c
-rwxr-xr-x 1 asus asus 16256 Oct 10 21:09 procfs_read
-rw-rw-r-- 1 asus asus 1175 Oct 10 21:08 procfs_read.c
-rwxr-xr-x 1 asus asus 16056 Oct 10 12:09 program
-rwxr-xr-x 1 asus asus 16056 Oct 10 16:42 program_dynamic
-rwxr-xr-x 1 asus asus 16024 Oct 10 16:48 program_shared
-rwxr-xr-x 1 asus asus 785400 Oct 10 16:42 program_static
drwxr-xr-x 2 asus asus 4096 Oct 13 21:17 test_dir
-rw-rw-r-- 1 asus asus 23 Oct 13 21:06 test_file.txt
asus@LAPTOP-BV04QHF1:~$ cd /tmp
asus@LAPTOP-BV04QHF1:/tmp$ mkdir test_dir
asus@LAPTOP-BV04QHF1:/tmp$ cd test_dir
```

- Pipelining

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ ps aux | grep bash | wc -l
3
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- Redirection

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ echo "Hello World" > output.txt
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ cat output.txt
Hello World
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ cat output.txt >> output.txt
cat: output.txt: input file is output file
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ cat < output.txt
Hello World
```

- Background processes

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ sleep 30 &
[1] 1005
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ jobs
[1]+  Running                  sleep 30 &
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ fg %1
sleep 30
```

2.2.2 Program C - Shell Sederhana

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nano simple_shell.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc -o simple_shell simple_shell.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ ./simple_shell
myshell> echo Hallo world
Hallo world
myshell> ^Z
[1]+  Stopped                  ./simple_shell
```

2.3 System Calls

System Calls adalah mekanisme yang digunakan program untuk berkomunikasi langsung dengan kernel sistem operasi. Dalam praktik ini, pengguna mempelajari tiga jenis system calls utama. Pertama, Process Control System Calls seperti `fork()`, `exec()`, dan `wait()` digunakan untuk membuat, menjalankan, dan mengelola proses. Kedua, File Management System Calls seperti `open()`, `write()`, `read()`, dan `close()` digunakan untuk mengelola file pada sistem. Ketiga, Information Maintenance System Calls seperti `getpid()`, `getuid()`, dan `time()` digunakan untuk memperoleh informasi tentang proses dan waktu sistem. Melalui praktik ini, pengguna memahami bagaimana sistem operasi menyediakan layanan penting untuk pengelolaan proses, file, dan informasi sistem secara efisien.

2.3.1 Process Control System Calls

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nano process_syscalls.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc -o process_syscalls process_syscalls.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 480
Parent waiting for child (PID: 481)...
Child PID: 481
Child's Parent PID: 480
Child executing 'ls' command...
total 44
-rwxr-xr-x 1 asus asus 16360 Oct 13 22:04 process_syscalls
-rw-r--r-- 1 asus asus 798 Oct 13 22:04 process_syscalls.c
-rwxr-xr-x 1 asus asus 16528 Oct 13 21:48 simple_shell
-rw-r--r-- 1 asus asus 1190 Oct 13 21:48 simple_shell.c
Child exited with status: 0
```

2.3.2 File Management System Calls

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nano file_syscalls.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc -o file_syscalls file_syscalls.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
```

2.3.3 Information Maintenance System Calls

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nano info_syscalls.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc -o info_syscalls info_syscalls.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ ./info_syscalls
=== Information Maintenance System Calls ===
Process ID: 509
Parent Process ID: 284
User ID: 1000
Group ID: 1000
Current time: Mon Oct 13 22:16:17 2025
Clock ticks since boot: 1718456236
Sleeping for 2 seconds...
Awake!
```

2.4 System Services

System Services di Linux adalah proses yang berjalan di latar belakang untuk menyediakan berbagai fungsi penting seperti jaringan, keamanan, dan manajemen sistem. Dalam praktik ini, digunakan systemd sebagai sistem inisialisasi modern untuk mengelola layanan. Dengan perintah seperti `systemctl list-units`, pengguna dapat melihat daftar layanan aktif, memeriksa statusnya menggunakan `systemctl status`, dan melihat log aktivitas melalui `journalctl`. Selain itu, dibuat program daemon sederhana (`my_daemon.c`) yang berjalan otomatis di latar belakang dan mencatat aktivitas ke `syslog`. Pembuatan file service di `/etc/systemd/system/` memungkinkan daemon tersebut dikelola oleh `systemd`. Praktik ini membantu memahami bagaimana sistem operasi menjalankan, memantau, dan mengontrol layanan secara efisien serta berkesinambungan.

2.4.1 Menggunakan system

- Lihat semua service

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ systemctl list-units --type=service
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
console-getty.service              loaded active running Console Getty
console-setup.service               loaded active exited Set console font and keypad
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
keyboard-setup.service              loaded active exited Set the console keyboard layout
kmod-static-nodes.service           loaded active exited Create List of Static Device Nodes
rsyslog.service                    loaded active running System Logging Service
setvtrgb.service                   loaded active exited Set console scheme
snapd.seeded.service               loaded active exited Wait until snapd is fully seeded
systemd-binfmt.service              loaded active exited Set Up Additional Binary Formats
systemd-journal-flush.service        loaded active exited Flush Journal to Persistent Storage
systemd-journald.service             loaded active running Journal Service
systemd-logind.service              loaded active running User Login Management
systemd-modules-load.service         loaded active exited Load Kernel Modules
systemd-remount-fs.service           loaded active exited Remount Root and Kernel File Systems
systemd-resolved.service             loaded active running Network Name Resolution
systemd-sysctl.service               loaded active exited Apply Kernel Variables
systemd-timesyncd.service            loaded active running Network Time Synchronization
systemd-tmpfiles-setup-dev-early.service loaded active exited Create Static Device Nodes in /dev gracefully
systemd-tmpfiles-setup-dev.service  loaded active exited Create Static Device Nodes in /dev
systemd-tmpfiles-setup.service       loaded active exited Create Volatile Files and Directories
systemd-udev-trigger.service         loaded active exited Coldplug All udev Devices
systemd-udevd.service               loaded active running Rule-based Manager for Device Events and Files
systemd-update-utmp.service          loaded active exited Record System Boot/Shutdown in UTMP
systemd-user-sessions.service        loaded active exited Permit User Sessions
unattended-upgrades.service          loaded active running Unattended Upgrades Shutdown
user-runtime-dir@1000.service        loaded active exited User Runtime Directory /run/user/1000
user@1000.service                   loaded active running User Manager for UID 1000
wsl-pro.service                     loaded active running Bridge to Ubuntu Pro agent on Windows

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
          ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
          SUB     → The low-level unit activation state, values depend on unit type.

30 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.
```

- Status service

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
```

- Lihat log service

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ journalctl -u ssh -n 20
Oct 09 18:55:00 LAPTOP-BV040HF1 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 09 18:55:00 LAPTOP-BV040HF1 sshd[3041]: Server listening on 0.0.0.0 port 22.
Oct 09 18:55:00 LAPTOP-BV040HF1 sshd[3041]: Server listening on :: port 22.
Oct 09 18:55:00 LAPTOP-BV040HF1 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```


○ Lihat service yang berjalan saat boot

```
asus@LAPTOP-BV04OHHF1:/tmp/test_dir$ systemctl list-unit-files --type=service --state=enabled
UNIT FILE                                STATE PRESET
apparmor.service                        enabled enabled
apport.service                          enabled enabled
cloud-config.service                    enabled enabled
cloud-init-local.service                 enabled enabled
cloud-init.service                       enabled enabled
console-setup.service                   enabled enabled
cron.service                            enabled enabled
dmesg.service                           enabled enabled
e2scrub_reap.service                     enabled enabled
getty@.service                           enabled enabled
keyboard-setup.service                   enabled enabled
landscape-client.service                 enabled enabled
networkd-dispatcher.service              enabled enabled
rsyslog.service                          enabled enabled
setvtrgb.service                         enabled enabled
snapd.apparmor.service                   enabled enabled
snapd.autoimport.service                 enabled enabled
snapd.core-fixup.service                 enabled enabled
snapd.recovery-chooser-trigger.service    enabled enabled
snapd.seeded.service                     enabled enabled
snapd.service                            enabled enabled
snapd.system-shutdown.service             enabled enabled
systemd-pstore.service                   enabled enabled
systemd-resolved.service                 enabled enabled
systemd-timesyncd.service                 enabled enabled
ua-reboot-cmds.service                   enabled enabled
ubuntu-advantage.service                 enabled enabled
unattended-upgrades.service               enabled enabled
wsl-pro.service                          enabled enabled

30 unit files listed.
```

2.4.2 Membuat Custom System Service

```
asus@LAPTOP-BV04OHHF1:/tmp/test_dir$ nano my_daemon.c
asus@LAPTOP-BV04OHHF1:/tmp/test_dir$ gcc -o my_daemon my_daemon.c
```

```
asus@LAPTOP-BV04OHHF1:~$ nano /etc/systemd/system/my_daemon.service
[asus] password for asus:
asus@LAPTOP-BV04OHHF1:/tmp/test_dir$ sudo nano /etc/systemd/system/my_daemon.service
asus@LAPTOP-BV04OHHF1:/tmp/test_dir$ sudo systemctl daemon-reload
asus@LAPTOP-BV04OHHF1:/tmp/test_dir$ sudo systemctl start my_daemon
Job for my_daemon.service failed because the control process exited with error code.
See "systemctl status my_daemon.service" and "journalctl -xe my_daemon.service" for details.
asus@LAPTOP-BV04OHHF1:/tmp/test_dir$ sudo systemctl status my_daemon
* my_daemon.service - My Custom Daemon
   Loaded: loaded (/etc/systemd/system/my_daemon.service; disabled; preset: enabled)
   Active: failed (result: exit-code) since Mon 2025-10-13 22:51:31 WIB; 5s ago
     Process: 650 ExecStart=/path/to/my_daemon (code=exited, status=203/EXEC)
    CPU: 2ms

Oct 13 22:51:31 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 13 22:51:31 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 13 22:51:31 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 13 22:51:31 LAPTOP-BV04OHHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
asus@LAPTOP-BV04OHHF1:/tmp/test_dir$ sudo journalctl -u my_daemon
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 2.
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:48:56 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 3.
Oct 09 19:48:57 LAPTOP-BV04OHHF1 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 19:48:57 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 19:48:57 LAPTOP-BV04OHHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:48:57 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 4.
Oct 09 19:48:57 LAPTOP-BV04OHHF1 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 09 19:48:57 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 09 19:48:57 LAPTOP-BV04OHHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:48:57 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 09 19:48:57 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 09 19:48:57 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:48:57 LAPTOP-BV04OHHF1 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 09 19:43:11 LAPTOP-BV04OHHF1 systemd[1]: Started my_daemon.service - My Custom Daemon.
Oct 09 19:43:11 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Main process exited, code=exited, status=203/EXEC
Oct 09 19:43:11 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:43:11 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 09 19:43:11 LAPTOP-BV04OHHF1 systemd[1]: Started my_daemon.service - My Custom Daemon.
Oct 09 19:43:11 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Main process exited, code=exited, status=203/EXEC
Oct 09 19:43:11 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:43:11 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 2.
Oct 09 19:43:12 LAPTOP-BV04OHHF1 systemd[1]: Started my_daemon.service - My Custom Daemon.
Oct 09 19:43:12 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Main process exited, code=exited, status=203/EXEC
Oct 09 19:43:12 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 09 19:43:12 LAPTOP-BV04OHHF1 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 3.
[22]~$
[22]~$ sudo journalctl -u my_daemon
asus@LAPTOP-BV04OHHF1:/tmp/test_dir$ sudo systemctl stop my_daemon
```

2.5 Linkers and Loaders

Pada Linux, *Linkers and Loaders* berperan penting dalam proses pembuatan program dari kode sumber menjadi file eksekusi. Linker bertugas menggabungkan beberapa file object (.o) seperti main.o dan math_ops.o menjadi satu program utuh, sedangkan loader memuat program tersebut ke memori saat dijalankan. Proses kompilasi dilakukan bertahap: preprocessing, compilation, assembly, dan linking. Dalam praktik ini, dibuat fungsi add() dan multiply() yang dihubungkan melalui file header math_ops.h. Selain itu, dibedakan antara *static linking* (semua library disertakan langsung dalam program) dan *dynamic linking* (library eksternal digunakan saat runtime). Shared library (libmathops.so) memungkinkan efisiensi memori dan kemudahan pemeliharaan karena dapat digunakan oleh banyak program sekaligus.

2.5.1 Proses Linking

- **Buat File math_ops.h**

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nano math_ops.h
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- **Buat File math_ops.c**

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nano math_ops.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- **Buat File main.c**

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nano main.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- **Preprocessing**

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc -E main.c -o main.i
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc -E math_ops.c -o math_ops.i
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- **Compilation (assembly)**

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc -S main.c -o main.s
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc -S math_ops.c -o math_ops.s
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- **Assembly (object files)**

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc -c main.c -o main.o
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc -c math_ops.c -o math_ops.o
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- **Linking**

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc main.o math_ops.o -o program
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- Lihat symbols dalam object file

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nm main.o
0000000000000000 T main
0000000000000000 U multiply
0000000000000000 U printf
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nm math_ops.o
0000000000000000 T add
0000000000000000 U multiply
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nm program
000000000003dc8 d _DYNAMIC
000000000003fbb d _GLOBAL_OFFSET_TABLE_
0000000000002000 R __IO_stdin_used
0000000000000000 w __ITM_deregisterTMCloneTable
0000000000000000 w __ITM_registerTMCloneTable
0000000000002140 r __FRAME_END__
000000000000201c r __GNU_EH_FRAME_HDR
0000000000004010 D __TMC_END__
000000000000038c r __abi_tag
00000000000004010 B __bss_start
00000000000004010 w __cxa_finalize@GLIBC_2.2.5
0000000000004000 D __data_start
0000000000001100 t __do_global_ctors_aux
0000000000003dc0 d __do_global_ctors_aux_fini_array_entry
0000000000004000 D __dso_handle
0000000000003db8 d __frame_dummy_init_array_entry
0000000000000000 w __gmon_start__
0000000000004010 D __edata
00000000000004010 B __end
00000000000011d4 T _fini
0000000000001000 T _init
0000000000001060 T _start
00000000000011a2 T add
00000000000004010 b completed.0
0000000000004000 W data_start
0000000000001090 t deregister_tm_clones
0000000000001140 t frame_dummy
0000000000001149 T main
00000000000011ba U multiply
00000000000010c0 t register_tm_clones
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- Lihat dependencies

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ ldd program
linux-vdso.so.1 (0x00007ffffa24e700)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000071e460c00000)
/lib64/ld-linux-x86-64.so.2 (0x000071e460f7c000)
```

2.5.2 Static vs Dynamic Linking

- Static linking

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc main.c math_ops.c -o program_static -static
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- Dynamic linking (default)

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc main.c math_ops.c -o program_dynamic
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- Bandingkan ukuran

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 asus asus 16K Oct 13 23:16 program_dynamic
-rwxr-xr-x 1 asus asus 767K Oct 13 23:15 program_static
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- Lihat dependencies

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 asus asus 16K Oct 13 23:16 program_dynamic
-rwxr-xr-x 1 asus asus 767K Oct 13 23:15 program_static
asus@LAPTOP-BV040HF1:/tmp/test_dir$ ldd program_static
not a dynamic executable
asus@LAPTOP-BV040HF1:/tmp/test_dir$ ldd program_dynamic
linux-vdso.so.1 (0x00007ffe6a94e000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000075503dc00000)
/lib64/ld-linux-x86-64.so.2 (0x000075503df42000)
asus@LAPTOP-BV040HF1:/tmp/test_dir$
```

2.5.3 Membuat dan Menggunakan Shared Library

- Compile sebagai shared library

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ gcc -c -fPIC math_ops.c -o math_ops.o
asus@LAPTOP-BV040HF1:/tmp/test_dir$ gcc -shared -o libmathops.so math_ops.o
asus@LAPTOP-BV040HF1:/tmp/test_dir$
```

- Compile program yang menggunakan library

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ gcc main.c -L. -lmathops -o program_shared
asus@LAPTOP-BV040HF1:/tmp/test_dir$
```

- Jalankan dengan LD_LIBRARY_PATH

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ LD_LIBRARY_PATH=. ./program_shared
5 + 3 = 8
5 * 3 = 15
asus@LAPTOP-BV040HF1:/tmp/test_dir$
```

- Lihat informasi loader

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ readelf -d program_shared
Dynamic section at offset 0x2da8 contains 28 entries:
  Tag                Type                             Name/Value
0x0000000000000001  (NEEDED)                         Shared library: [libmathops.so]
0x0000000000000001  (NEEDED)                         Shared library: [libc.so.6]
0x000000000000000c  (INIT)                           0x1000
0x000000000000000d  (FINI)                           0x11e4
0x0000000000000019  (INIT_ARRAY)                     0x3d98
0x000000000000001b  (INIT_ARRAYSZ)                   8 (bytes)
0x000000000000001a  (INIT_ARRAY)                     0x3da0
0x000000000000001c  (FINI_ARRAYSZ)                   8 (bytes)
0x00000000000000f5  (GNU_HASH)                       0x3b0
0x0000000000000005  (STRTAB)                         0x4b0
0x0000000000000006  (SYMTAB)                         0x3d8
0x000000000000000a  (STRSZ)                          170 (bytes)
0x000000000000000b  (SYMMENT)                        24 (bytes)
0x0000000000000015  (DEBUG)                          0x0
0x0000000000000002  (PLTGO)                          0x3fa8
0x0000000000000002  (PLTRELSZ)                       72 (bytes)
0x0000000000000014  (PLTREL)                         RELA
0x0000000000000017  (JMPREL)                         0x60
0x0000000000000007  (RELA)                           0x5a0
0x0000000000000008  (RELA_SZ)                        192 (bytes)
0x0000000000000009  (RELA_ENT)                       24 (bytes)
0x000000000000001e  (FLAGS)                          BIND_NOW
0x00000000000000fb  (FLAGS_1)                        Flags: NOW PIE
0x00000000000000ff  (VERNEED)                        0x570
0x00000000000000ff  (VERNEEDNUM)                     1
0x00000000000000ff  (VERSYM)                         0x55a
0x00000000000000ff  (RELACOUNT)                      3
0x0000000000000000  (NULL)                           0x0
asus@LAPTOP-BV040HF1:/tmp/test_dir$
```

2.6 Why Applications Are Operating-System Specific

Aplikasi bersifat spesifik terhadap sistem operasi karena setiap OS memiliki *system call*, format biner, dan API (Application Programming Interface) yang berbeda. Pada praktik ini, program `os_specific.c` menunjukkan bagaimana kode C harus disesuaikan untuk Linux dan Windows. Misalnya, di Linux digunakan pustaka `<unistd.h>` dan fungsi `uname()` untuk mendapatkan informasi sistem, sedangkan di Windows menggunakan `<windows.h>`. Selain itu, perbedaan format biner seperti ELF di Linux dan PE di Windows menyebabkan program tidak dapat dijalankan lintas sistem. Dengan perintah seperti `readelf` dan `objdump`, pengguna dapat menganalisis struktur file eksekusi dan memahami bagaimana sistem operasi memuat serta menjalankan aplikasi tersebut.

2.6.1 System Call Differences

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nano os_specific.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

2.6.2 Binary Format Analysis

- Lihat format executable

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=0c5145e0b63b47095af229368edc342f23876fc2, for GNU/Linux 3.2.0, not stripped
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- Lihat header ELF

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ readelf -h program
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:             ELF64
  Data:              2's complement, little endian
  Version:           1 (current)
  OS/ABI:            UNIX - System V
  ABI Version:       0
  Type:              DYN (Position-Independent Executable file)
  Machine:           Advanced Micro Devices X86-64
  Version:           0x1
  Entry point address: 0x1060
  Start of program headers: 64 (bytes into file)
  Start of section headers: 14872 (bytes into file)
  Flags:             0x0
  Size of this header: 64 (bytes)
  Size of program headers: 56 (bytes)
  Number of program headers: 13
  Size of section headers: 64 (bytes)
  Number of section headers: 31
  Section header string table index: 30
asus@LAPTOP-BV04QHF1:/tmp/test_dir$
```

- Lihat section headers

```

asus@LAPTOP-BV04OHFI:/tmp$ readelf -X program
There are 31 section headers, starting at offset 0x1000:

Section Headers:
  [Nr] Name              Type              Address            Offset
     Size              Info              File Flags      Link Defn
  [ 0]                     NULL              0000000000000000  0
  [ 1] .interp               PROGBITS          0000000000000010  0
  [ 2] .note.gnu.pr[...]     NOTE              0000000000000018  0
  [ 3] .note.gnu.bu[...]     NOTE              000000000000001d  0
  [ 4] .note.ABI-tag         NOTE              000000000000001e  0
  [ 5] .gnu.hash             GNU_HASH          000000000000001f  0
  [ 6] .dynsym               DWYNSYM          0000000000000020  0
  [ 7] .dynstr               STRTAB           0000000000000021  0
  [ 8] .gnu.version          VERSN            0000000000000022  0
  [ 9] .gnu.version_r        VERSN            0000000000000023  0
  [10] .rela.dyn              RELA              0000000000000024  0
  [11] .rela.plt              RELA              0000000000000025  0
  [12] .reloc                 PROGBITS          0000000000000026  0
  [13] .plt                   PROGBITS          0000000000000027  0
  [14] .plt.got               PROGBITS          0000000000000028  0
  [15] .plt.sec               PROGBITS          0000000000000029  0
  [16] .text                  PROGBITS          000000000000002a  0
  [17] .fini                  PROGBITS          000000000000002b  0
  [18] .rodata                PROGBITS          000000000000002c  0
  [19] .eh_frame_hdr          PROGBITS          000000000000002d  0
  [20] .eh_frame              PROGBITS          000000000000002e  0
  [21] .init_array            INIT_ARRAY        000000000000002f  0
  [22] .fini_array            FINI_ARRAY        0000000000000030  0
  [23] .dynamic                DYNAMIC           0000000000000031  0
  [24] .got                   PROGBITS          0000000000000032  0
  [25] .data                  PROGBITS          0000000000000033  0
  [26] .bss                   PROGBITS          0000000000000034  0
  [27] .common                PROGBITS          0000000000000035  0
  [28] .rela.dyn              RELA              0000000000000036  0
  [29] .rela.plt              RELA              0000000000000037  0
  [30] .reloc                 PROGBITS          0000000000000038  0
  [31] .eh_frame_hdr          PROGBITS          0000000000000039  0
  [32] .eh_frame              PROGBITS          000000000000003a  0
  [33] .init_array            INIT_ARRAY        000000000000003b  0
  [34] .fini_array            FINI_ARRAY        000000000000003c  0
  [35] .dynamic                DYNAMIC           000000000000003d  0
  [36] .got                   PROGBITS          000000000000003e  0
  [37] .data                  PROGBITS          000000000000003f  0
  [38] .bss                   PROGBITS          0000000000000040  0
  [39] .common                PROGBITS          0000000000000041  0
  [40] .rela.dyn              RELA              0000000000000042  0
  [41] .rela.plt              RELA              0000000000000043  0
  [42] .reloc                 PROGBITS          0000000000000044  0
  [43] .eh_frame_hdr          PROGBITS          0000000000000045  0
  [44] .eh_frame              PROGBITS          0000000000000046  0
  [45] .init_array            INIT_ARRAY        0000000000000047  0
  [46] .fini_array            FINI_ARRAY        0000000000000048  0
  [47] .dynamic                DYNAMIC           0000000000000049  0
  [48] .got                   PROGBITS          000000000000004a  0
  [49] .data                  PROGBITS          000000000000004b  0
  [50] .bss                   PROGBITS          000000000000004c  0
  [51] .common                PROGBITS          000000000000004d  0
  [52] .rela.dyn              RELA              000000000000004e  0
  [53] .rela.plt              RELA              000000000000004f  0
  [54] .reloc                 PROGBITS          0000000000000050  0
  [55] .eh_frame_hdr          PROGBITS          0000000000000051  0
  [56] .eh_frame              PROGBITS          0000000000000052  0
  [57] .init_array            INIT_ARRAY        0000000000000053  0
  [58] .fini_array            FINI_ARRAY        0000000000000054  0
  [59] .dynamic                DYNAMIC           0000000000000055  0
  [60] .got                   PROGBITS          0000000000000056  0
  [61] .data                  PROGBITS          0000000000000057  0
  [62] .bss                   PROGBITS          0000000000000058  0
  [63] .common                PROGBITS          0000000000000059  0
  [64] .rela.dyn              RELA              000000000000005a  0
  [65] .rela.plt              RELA              000000000000005b  0
  [66] .reloc                 PROGBITS          000000000000005c  0
  [67] .eh_frame_hdr          PROGBITS          000000000000005d  0
  [68] .eh_frame              PROGBITS          000000000000005e  0
  [69] .init_array            INIT_ARRAY        000000000000005f  0
  [70] .fini_array            FINI_ARRAY        0000000000000060  0
  [71] .dynamic                DYNAMIC           0000000000000061  0
  [72] .got                   PROGBITS          0000000000000062  0
  [73] .data                  PROGBITS          0000000000000063  0
  [74] .bss                   PROGBITS          0000000000000064  0
  [75] .common                PROGBITS          0000000000000065  0
  [76] .rela.dyn              RELA              0000000000000066  0
  [77] .rela.plt              RELA              0000000000000067  0
  [78] .reloc                 PROGBITS          0000000000000068  0
  [79] .eh_frame_hdr          PROGBITS          0000000000000069  0
  [80] .eh_frame              PROGBITS          000000000000006a  0
  [81] .init_array            INIT_ARRAY        000000000000006b  0
  [82] .fini_array            FINI_ARRAY        000000000000006c  0
  [83] .dynamic                DYNAMIC           000000000000006d  0
  [84] .got                   PROGBITS          000000000000006e  0
  [85] .data                  PROGBITS          000000000000006f  0
  [86] .bss                   PROGBITS          0000000000000070  0
  [87] .common                PROGBITS          0000000000000071  0
  [88] .rela.dyn              RELA              0000000000000072  0
  [89] .rela.plt              RELA              0000000000000073  0
  [90] .reloc                 PROGBITS          0000000000000074  0
  [91] .eh_frame_hdr          PROGBITS          0000000000000075  0
  [92] .eh_frame              PROGBITS          0000000000000076  0
  [93] .init_array            INIT_ARRAY        0000000000000077  0
  [94] .fini_array            FINI_ARRAY        0000000000000078  0
  [95] .dynamic                DYNAMIC           0000000000000079  0
  [96] .got                   PROGBITS          000000000000007a  0
  [97] .data                  PROGBITS          000000000000007b  0
  [98] .bss                   PROGBITS          000000000000007c  0
  [99] .common                PROGBITS          000000000000007d  0
  [100] .rela.dyn              RELA              000000000000007e  0
  [101] .rela.plt              RELA              000000000000007f  0
  [102] .reloc                 PROGBITS          0000000000000080  0
  [103] .eh_frame_hdr          PROGBITS          0000000000000081  0
  [104] .eh_frame              PROGBITS          0000000000000082  0
  [105] .init_array            INIT_ARRAY        0000000000000083  0
  [106] .fini_array            FINI_ARRAY        0000000000000084  0
  [107] .dynamic                DYNAMIC           0000000000000085  0
  [108] .got                   PROGBITS          0000000000000086  0
  [109] .data                  PROGBITS          0000000000000087  0
  [110] .bss                   PROGBITS          0000000000000088  0
  [111] .common                PROGBITS          0000000000000089  0
  [112] .rela.dyn              RELA              000000000000008a  0
  [113] .rela.plt              RELA              000000000000008b  0
  [114] .reloc                 PROGBITS          000000000000008c  0
  [115] .eh_frame_hdr          PROGBITS          000000000000008d  0
  [116] .eh_frame              PROGBITS          000000000000008e  0
  [117] .init_array            INIT_ARRAY        000000000000008f  0
  [118] .fini_array            FINI_ARRAY        0000000000000090  0
  [119] .dynamic                DYNAMIC           0000000000000091  0
  [120] .got                   PROGBITS          0000000000000092  0
  [121] .data                  PROGBITS          0000000000000093  0
  [122] .bss                   PROGBITS          0000000000000094  0
  [123] .common                PROGBITS          0000000000000095  0
  [124] .rela.dyn              RELA              0000000000000096  0
  [125] .rela.plt              RELA              0000000000000097  0
  [126] .reloc                 PROGBITS          0000000000000098  0
  [127] .eh_frame_hdr          PROGBITS          0000000000000099  0
  [128] .eh_frame              PROGBITS          000000000000009a  0
  [129] .init_array            INIT_ARRAY        000000000000009b  0
  [130] .fini_array            FINI_ARRAY        000000000000009c  0
  [131] .dynamic                DYNAMIC           000000000000009d  0
  [132] .got                   PROGBITS          000000000000009e  0
  [133] .data                  PROGBITS          000000000000009f  0
  [134] .bss                   PROGBITS          00000000000000a0  0
  [135] .common                PROGBITS          00000000000000a1  0
  [136] .rela.dyn              RELA              00000000000000a2  0
  [137] .rela.plt              RELA              00000000000000a3  0
  [138] .reloc                 PROGBITS          00000000000000a4  0
  [139] .eh_frame_hdr          PROGBITS          00000000000000a5  0
  [140] .eh_frame              PROGBITS          00000000000000a6  0
  [141] .init_array            INIT_ARRAY        00000000000000a7  0
  [142] .fini_array            FINI_ARRAY        00000000000000a8  0
  [143] .dynamic                DYNAMIC           00000000000000a9  0
  [144] .got                   PROGBITS          00000000000000aa  0
  [145] .data                  PROGBITS          00000000000000ab  0
  [146] .bss                   PROGBITS          00000000000000ac  0
  [147] .common                PROGBITS          00000000000000ad  0
  [148] .rela.dyn              RELA              00000000000000ae  0
  [149] .rela.plt              RELA              00000000000000af  0
  [150] .reloc                 PROGBITS          00000000000000b0  0
  [151] .eh_frame_hdr          PROGBITS          00000000000000b1  0
  [152] .eh_frame              PROGBITS          00000000000000b2  0
  [153] .init_array            INIT_ARRAY        00000000000000b3  0
  [154] .fini_array            FINI_ARRAY        00000000000000b4  0
  [155] .dynamic                DYNAMIC           00000000000000b5  0
  [156] .got                   PROGBITS          00000000000000b6  0
  [157] .data                  PROGBITS          00000000000000b7  0
  [158] .bss                   PROGBITS          00000000000000b8  0
  [159] .common                PROGBITS          00000000000000b9  0
  [160] .rela.dyn              RELA              00000000000000ba  0
  [161] .rela.plt              RELA              00000000000000bb  0
  [162] .reloc                 PROGBITS          00000000000000bc  0
  [163] .eh_frame_hdr          PROGBITS          00000000000000bd  0
  [164] .eh_frame              PROGBITS          00000000000000be  0
  [165] .init_array            INIT_ARRAY        00000000000000bf  0
  [166] .fini_array            FINI_ARRAY        00000000000000c0  0
  [167] .dynamic                DYNAMIC           00000000000000c1  0
  [168] .got                   PROGBITS          00000000000000c2  0
  [169] .data                  PROGBITS          00000000000000c3  0
  [170] .bss                   PROGBITS          00000000000000c4  0
  [171] .common                PROGBITS          00000000000000c5  0
  [172] .rela.dyn              RELA              00000000000000c6  0
  [173] .rela.plt              RELA              00000000000000c7  0
  [174] .reloc                 PROGBITS          00000000000000c8  0
  [175] .eh_frame_hdr          PROGBITS          00000000000000c9  0
  [176] .eh_frame              PROGBITS          00000000000000ca  0
  [177] .init_array            INIT_ARRAY        00000000000000cb  0
  [178] .fini_array            FINI_ARRAY        00000000000000cc  0
  [179] .dynamic                DYNAMIC           00000000000000cd  0
  [180] .got                   PROGBITS          00000000000000ce  0
  [181] .data                  PROGBITS          00000000000000cf  0
  [182] .bss                   PROGBITS          00000000000000d0  0
  [183] .common                PROGBITS          00000000000000d1  0
  [184] .rela.dyn              RELA              00000000000000d2  0
  [185] .rela.plt              RELA              00000000000000d3  0
  [186] .reloc                 PROGBITS          00000000000000d4  0
  [187] .eh_frame_hdr          PROGBITS          00000000000000d5  0
  [188] .eh_frame              PROGBITS          00000000000000d6  0
  [189] .init_array            INIT_ARRAY        00000000000000d7  0
  [190] .fini_array            FINI_ARRAY        00000000000000d8  0
  [191] .dynamic                DYNAMIC           00000000000000d9  0
  [192] .got                   PROGBITS          00000000000000da  0
  [193] .data                  PROGBITS          00000000000000db  0
  [194] .bss                   PROGBITS          00000000000000dc  0
  [195] .common                PROGBITS          00000000000000dd  0
  [196] .rela.dyn              RELA              00000000000000de  0
  [197] .rela.plt              RELA              00000000000000df  0
  [198] .reloc                 PROGBITS          00000000000000e0  0
  [199] .eh_frame_hdr          PROGBITS          00000000000000e1  0
  [200] .eh_frame              PROGBITS          00000000000000e2  0
  [201] .init_array            INIT_ARRAY        00000000000000e3  0
  [202] .fini_array            FINI_ARRAY        00000000000000e4  0
  [203] .dynamic                DYNAMIC           00000000000000e5  0
  [204] .got                   PROGBITS          00000000000000e6  0
  [205] .data                  PROGBITS          00000000000000e7  0
  [206] .bss                   PROGBITS          00000000000000e8  0
  [207] .common                PROGBITS          00000000000000e9  0
  [208] .rela.dyn              RELA              00000000000000ea  0
  [209] .rela.plt              RELA              00000000000000eb  0
  [210] .reloc                 PROGBITS          00000000000000ec  0
  [211] .eh_frame_hdr          PROGBITS          00000000000000ed  0
  [212] .eh_frame              PROGBITS          00000000000000ee  0
  [213] .init_array            INIT_ARRAY        00000000000000ef  0
  [214] .fini_array            FINI_ARRAY        00000000000000f0  0
  [215] .dynamic                DYNAMIC           00000000000000f1  0
  [216] .got                   PROGBITS          00000000000000f2  0
  [217] .data                  PROGBITS          00000000000000f3  0
  [218] .bss                   PROGBITS          00000000000000f4  0
  [219] .common                PROGBITS          00000000000000f5  0
  [220] .rela.dyn              RELA              00000000000000f6  0
  [221] .rela.plt              RELA              00000000000000f7  0
  [222] .reloc                 PROGBITS          00000000000000f8  0
  [223] .eh_frame_hdr          PROGBITS          00000000000000f9  0
  [224] .eh_frame              PROGBITS          00000000000000fa  0
  [225] .init_array            INIT_ARRAY        00000000000000fb  0
  [226] .fini_array            FINI_ARRAY        00000000000000fc  0
  [227] .dynamic                DYNAMIC           00000000000000fd  0
  [228] .got                   PROGBITS          00000000000000fe  0
  [229] .data                  PROGBITS          00000000000000ff  0
  [230] .bss                   PROGBITS          0000000000000100  0
  [231] .common                PROGBITS          0000000000000101  0
  [232] .rela.dyn              RELA              0000000000000102  0
  [233] .rela.plt              RELA              0000000000000103  0
  [234] .reloc                 PROGBITS          0000000000000104  0
  [235] .eh_frame_hdr          PROGBITS          0000000000000105  0
  [236] .eh_frame              PROGBITS          0000000000000106  0
  [237] .init_array            INIT_ARRAY        0000000000000107  0
  [238] .fini_array            FINI_ARRAY        0000000000000108  0
  [239] .dynamic                DYNAMIC           0000000000000109  0
  [240] .got                   PROGBITS          000000000000010a  0
  [241] .data                  PROGBITS          000000000000010b  0
  [242] .bss                   PROGBITS          000000000000010c  0
  [243] .common                PROGBITS          000000000000010d  0
  [244] .rela.dyn              RELA              000000000000010e  0
  [245] .rela.plt              RELA              000000000000010f  0
  [246] .reloc                 PROGBITS          0000000000000110  0
  [247] .eh_frame_hdr          PROGBITS          0000000000000111  0
  [248] .eh_frame              PROGBITS          0000000000000112  0
  [249] .init_array            INIT_ARRAY        0000000000000113  0
  [250] .fini_array            FINI_ARRAY        0000000000000114  0
  [251] .dynamic                DYNAMIC           0000000000000115  0
  [252] .got                   PROGBITS          0000000000000116  0
  [253] .data                  PROGBITS          0000000000000117  0
  [254] .bss                   PROGBITS          0000000000000118  0
  [255] .common                PROGBITS          0000000000000119  0
  [256] .rela.dyn              RELA              000000000000011a  0
  [257] .rela.plt              RELA              000000000000011b  0
  [258] .reloc                 PROGBITS          000000000000011c  0
  [259] .eh_frame_hdr          PROGBITS          000000000000011d  0
  [260] .eh_frame              PROGBITS          000000000000011e  0
  [261] .init_array            INIT_ARRAY        000000000000011f  0
  [262] .fini_array            FINI_ARRAY        0000000000000120  0
  [263] .dynamic                DYNAMIC           0000000000000121  0
  [264] .got                   PROGBITS          0000000000000122  0
  [265] .data                  PROGBITS          0000000000000123  0
  [266] .bss                   PROGBITS          0000000000000124  0
  [267] .common                PROGBITS          0000000000000125  0
  [268] .rela.dyn              RELA              0000000000000126  0
  [269] .rela.plt              RELA              0000000000000127  0
  [270] .reloc                 PROGBITS          0000000000000128  0
  [271] .eh_frame_hdr          PROGBITS          0000000000000129  0
  [272] .eh_frame              PROGBITS          000000000000012a  0
  [273] .init_array            INIT_ARRAY        000000000000012b  0
  [274] .fini_array            FINI_ARRAY        000000000000012c  0
  [275] .dynamic                DYNAMIC           000000000000012d  0
  [276] .got                   PROGBITS          000000000000012e  0
  [277] .data                  PROGBITS          000000000000012f  0
  [278] .bss                   PROGBITS          0000000000000130  0
  [279] .common                PROGBITS          0000000000000131  0
  [280] .rela.dyn              RELA              0000000000000132  0
  [281] .rela.plt              RELA              0000000000000133  0
  [282] .reloc                 PROGBITS          0000000000000134  0
  [283] .eh_frame_hdr          PROGBITS          0000000000000135  0
  [284] .eh_frame              PROGBITS          0000000000000136  0
  [285] .init_array            INIT_ARRAY        0000000000000137  0
  [286] .fini_array            FINI_ARRAY        0000000000000138  0
  [287] .dynamic                DYNAMIC           0000000000000139  0
  [288] .got                   PROGBITS          000000000000013a  0
  [289] .data                  PROGBITS          000000000000013b  0
  [290] .bss                   PROGBITS          000000000000013c  0
  [291] .common                PROGBITS          000000000000013d  0
  [292] .rela.dyn              RELA              000000000000013e  0
  [293] .rela.plt              RELA              000000000000013f  0
  [294] .reloc                 PROGBITS          0000000000000140  0
  [295] .eh_frame_hdr          PROGBITS          0000000000000141  0
  [296] .eh_frame              PROGBITS          0000000000000142  0
  [297] .init_array            INIT_ARRAY        0000000000000143  0
  [298] .fini_array            FINI_ARRAY        0000000000000144  0
  [299] .dynamic                DYNAMIC           0000000000000145  0
  [300] .got                   PROGBITS          0000000000000146  0
  [301] .data                  PROGBITS          0000000000000147  0
  [302] .bss                   PROGBITS          0000000000000148  0
  [303] .common                PROGBITS          0000000000000149  0
  [304] .rela.dyn              RELA              000000000000014a  0
  [305] .rela.plt              RELA              000000000000014b  0
  [306] .reloc                 PROGBITS          000000000000014c  0
  [307] .eh_frame_hdr          PROGBITS          000000000000014d  0
  [308] .eh_frame              PROGBITS          000000000000014e  0
  [309] .init_array            INIT_ARRAY        000000000000014f  0
  [310] .fini_array            FINI_ARRAY        0000000000000150  0
  [311] .dynamic                DYNAMIC           0000000000000151  0
  [312] .got                   PROGBITS          0000000000000152  0
  [313] .data                  PROGBITS          0000000000000153  0
  [314] .bss                   PROGBITS          0000000000000154  0
  [315] .common                PROGBITS          0000000000000155  0
  [316] .rela.dyn              RELA              0000000000000156  0
  [317] .rela.plt              RELA              0000000000000157  0
  [318] .reloc                 PROGBITS          0000000000000158  0
  [319] .eh_frame_hdr          PROGBITS          0000000000000159  0
  [320] .eh_frame              PROGBITS          000000000000015a  0
  [321] .init_array            INIT_ARRAY        000000000000015b  0
  [322] .fini_array            FINI_ARRAY        000000000000015c  0
  [323] .dynamic                DYNAMIC           000000000000015d  0
  [324] .got                   PROGBITS          000000000000015e  0
  [325] .data                  PROGBITS          000000000000015f  0
  [326] .bss                   PROGBITS          0000000000000160  0
  [327] .common                PROGBITS          0000000000000161  0
  [328] .rela.dyn              RELA              0000000000000162  0
  [329] .rela.plt              RELA              0000000000000163  0
  [330] .reloc                 PROGBITS          0000000000000164  0
  [331] .eh_frame_hdr          PROGBITS          0000000000000165  0
  [332] .eh_frame              PROGBITS          0000000000000166  0
  [333] .init_array            INIT_ARRAY        0000000000000167  0
  [334] .fini_array            FINI_ARRAY        0000000000000168  0
  [335] .dynamic                DYNAMIC           0000000000000169  0
  [336] .got                   PROGBITS          000000000000016a  0
  [337] .data                  PROGBITS          000000000000016b  0
  [338] .bss                   PROGBITS          000000000000016c  0
  [339] .common                PROGBITS          000000000000016d  0
  [340] .rela.dyn              RELA              000000000000016e  0
  [341] .rela.plt              RELA              000000000000016f  0
  [342] .reloc                 PROGBITS          0000000000000170  0
  [343] .eh_frame_hdr          PROGBITS          0000000000000171  0
  [344] .eh_frame              PROGBITS          0000000000000172  0
  [345] .init_array            INIT_ARRAY        0000000000000173  0
  [346] .fini_array            FINI_ARRAY        0000000000000174  0
  [347] .dynamic                DYNAMIC           0000000000000175  0
  [348] .got                   PROGBITS          0000000000000176  0
  [349] .data                  PROGBITS          0000000000000177  0
  [350] .bss                   PROGBITS          0000000000000178  0
  [351] .common                PROGBITS          0000000000000179  0
  [352] .rela.dyn              RELA              000000000000017a  0
  [353] .rela.plt              RELA              000000000000017b  0
  [354] .reloc                 PROGBITS          000000000000017c  0
  [355]
```

- Disassemble

```
asus@LAPTOP-BV04QHFI:/tmp/test_dir$ objdump -d program | head -50
program:      file format elf64-x86_64

Disassembly of section .init:

0000000000001000 <.init>:
1000: f3 0f 1e fa          endbr64
1004: 48 83 ec 08          sub    $0x8,%rsp
1008: 48 8b 05 d9 2f 00 00 mov    0x2fd9(<rip>),%rax    # 3fe8 <__gmon_start__@Base>
100c: 48 83 c0             test   %rax,%rax
1010: 74 02              je     .L016 <.init+0x16>
1012: ff 00              call   *(%rax,%rax)
1016: 48 83 c4 08          add    $0x8,%rsp
101a: c3                ret

Disassembly of section .plt:

0000000000001020 <.plt>:
1020: ff 25 9a 2f 00 00    push  0x2fa(<rip>)          # 3fc8 <_GLOBAL_OFFSET_TABLE_+0x8>
1024: ff 25 9c 2f 00 00    jmp    *0x2fc(<rip>)        # 3fc8 <_GLOBAL_OFFSET_TABLE_+0x10>
1028: 0f 1f 40 00          nopl   0x0(<rax>)
102c: f3 0f 1e fa          endbr64
1030: 00 00 00 00 00 00    push  $0x0
1034: c9 e2 ff ff ff      jmp    1028 <.init+0x28>
1038: 00 00              xchg   %ax,%ax

Disassembly of section .plt.got:

0000000000001040 <_cxa_finalize@plt>:
1040: f3 0f 1e fa          endbr64
1044: ff 25 ae 2f 00 00    jmp    *0x2fae(<rip>)       # 3ffa <__cxa_finalize@GLIBC_2.2.5>
1048: 66 0f 1f 40 00 00    nopl   0x0(<rax,%rax,1>)

Disassembly of section .plt.sec:

0000000000001060 <.print@plt>:
1060: f3 0f 1e fa          endbr64
1064: ff 25 76 2f 00 00    jmp    *0x2f76(<rip>)       # 3fd8 <.print@GLIBC_2.2.5>
1068: 66 0f 1f 40 00 00    nopl   0x0(<rax,%rax,1>)

Disassembly of section .text:

0000000000001080 <_start>:
1080: f3 0f 1e fa          endbr64
1084: 31 c3              xor    %ebp,%ebp
1088: 41 00 d1          mov    %rcx,%rcx
108c: 5e              pop    %rsi
1090: 48 89 e2          mov    %rsp,%rdi
1094: 48 83 e0 40       and    $0xfffffffffff0,%rsp
asus@LAPTOP-BV04QHFI:/tmp/test_dir$
```

2.7 OS Design and Implementation

Pada praktik ini, mahasiswa mempelajari bagaimana desain dan implementasi sistem operasi dilakukan melalui pembuatan *kernel module* sederhana di Linux. Modul kernel adalah komponen yang dapat dimuat dan dilepas dari kernel secara dinamis tanpa perlu melakukan reboot sistem. Pada file `hello_module.c`, digunakan fungsi `printk()` untuk menampilkan pesan ke *kernel log* saat modul dimuat (`insmod`) dan dilepas (`rmmod`). Proses kompilasi dilakukan menggunakan *Makefile* yang memanggil build system kernel melalui `make -C /lib/modules/....` Selain itu, perintah seperti `lsmod`, `dmesg`, dan `modinfo` digunakan untuk memantau aktivitas kernel dan modul yang aktif. Praktik ini membantu memahami struktur internal kernel serta interaksi antara modul dan sistem operasi.

2.7.1 Kernel Module Sederhana

```
asus@LAPTOP-BV04QHFI:/tmp/test_dir$ nano hello_module.c
asus@LAPTOP-BV04QHFI:/tmp/test_dir$
```

- Makefile

```
asus@LAPTOP-BV04QHFI:/tmp/test_dir$ sudo nano Makefile
```

- **Compile and Load**

```
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$ make
make -C /lib/modules/6.6.87.2-microsoft-standard-WSL2/build M=/tmp/test_dir modules
make[1]: *** /lib/modules/6.6.87.2-microsoft-standard-WSL2/build: No such file or directory. Stop.
make: *** [Makefile:4: all] Error 2
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$ sudo insmod hello_module.ko
insmod: ERROR: could not load module hello_module.ko: No such file or directory
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$ lsmod | grep hello
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$ dmesg | tail
[ 3158.145483] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.146833] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.146903] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.146916] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.147556] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.148886] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.258486] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3151.986230] systemd-journald[44]: /var/log/journal/0d8d8415eab84a3c92726d302daefa8c/user-1000-journal: Journal file uses a different sequence number ID, rotating.
[ 5422.849537] hv_utils: TimeSync IC version 4.0
[ 5438.781796] WSL (262) ERROR: CheckConnection: getaddrinfo() failed: -5
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$ sudo rmmod hello_module
rmmod: ERROR: Module hello_module is not currently loaded
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$ dmesg | tail
[ 3158.145483] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.146833] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.146903] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.146916] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.147556] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.148886] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3158.258486] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3151.986230] systemd-journald[44]: /var/log/journal/0d8d8415eab84a3c92726d302daefa8c/user-1000-journal: Journal file uses a different sequence number ID, rotating.
[ 5422.849537] hv_utils: TimeSync IC version 4.0
[ 5438.781796] WSL (262) ERROR: CheckConnection: getaddrinfo() failed: -5
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$
```

2.7.2 Monitoring Kernel

- **Lihat informasi kernel**

```
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$ uname -r
6.6.87.2-microsoft-standard-WSL2
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad540) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$
```

- **Lihat parameter kernel**

```
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$ sysctl -a | head -20
sysctl: permission denied on key 'kernel.cad_pid'
abi.vsyscall32 = 1
debug.exception-trace = 1
debug.kprobes-optimization = 1
dev.cdrom.autoclose = 1
dev.cdrom.autoeject = 0
dev.cdrom.check_media = 0
dev.cdrom.debug = 0
dev.cdrom.info = CD-ROM information, Id: cdrom.c 3.20 2803/12/17
dev.cdrom.info =
dev.cdrom.info = drive name:
dev.cdrom.info = drive speed:
dev.cdrom.info = drive # of slots:
dev.cdrom.info = Can close tray:
dev.cdrom.info = Can open tray:
dev.cdrom.info = Can lock tray:
dev.cdrom.info = Can change speed:
dev.cdrom.info = Can select disk:
dev.cdrom.info = Can read multisession:
dev.cdrom.info = Can read MCN:
dev.cdrom.info = Reports media changed:
```

- **Lihat kernel ring buffer**

```
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$ dmesg | tail -30
[ 2171.987860] mnt_init (247): drop_caches: 1
[ 3115.244650] Exception:
[ 3115.244685] Operation canceled @p910.cpp:258 (AcceptAsync)

[ 3115.406349] EXT4-fs (sdd): unmounting filesystem Scd5b39b-9668-41b0-92a5-b98ca83e543e.
[ 3149.163856] EXT4-fs (sdd): mounted filesystem Scd5b39b-9668-41b0-92a5-b98ca83e543e r/w with ordered data mode. Quota mode: none.
[ 3149.731730] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3149.734886] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3149.737637] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3149.738195] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3149.738720] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3149.957523] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.021830] systemd-journald[44]: Collecting audit messages is disabled.
[ 3150.093557] systemd-journald[44]: Received client request to flush runtime journal.
[ 3150.095580] systemd-journald[44]: File /var/log/journal/0d8d8415eab84a3c92726d302daefa8c/system-journal corrupted or uncleanly shut down, renaming and replacing.
[ 3150.142712] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.143541] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.144891] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.144916] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.144941] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.145451] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.144934] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.145483] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146833] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146449] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.146916] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.147556] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.148886] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3150.258486] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3151.986230] systemd-journald[44]: /var/log/journal/0d8d8415eab84a3c92726d302daefa8c/user-1000-journal: Journal file uses a different sequence number ID, rotating.
[ 5422.849537] hv_utils: TimeSync IC version 4.0
[ 5438.781796] WSL (262) ERROR: CheckConnection: getaddrinfo() failed: -5
asus@LAPTOP-BVBH0HFI:/tmp/test_dir$
```


- **Lihat kernel modules**

```
ksus@LAPTOP-BV8H0HF1:/tmp/test_dir$ lsmod
Module                  Size  Used By
tls                     98384  0
intel_rapl_msr          16384  0
intel_rapl_common       27768  1 intel_rapl_msr
kvm_and                 126976  0
kvm                     976752  1 kvm_and
irqbypass              12288  1 kvm
crc32c_intel            16384  2
battery                 20480  0
ac                      16384  0
sch_fq_codel            16384  1
configfs                53248  1
autofs4                 48864  0
br_netfilter            28672  0
bridge                  282024  1 br_netfilter
stp                     12288  1 bridge
tll                     12288  2 bridge, stp
ip_tables               28672  0
tun                     53248  0
ksus@LAPTOP-BV8H0HF1:/tmp/test_dir$ |
```

- **Informasi modul tertentu**

```
ksus@LAPTOP-BV8H0HF1:/tmp/test_dir$ modinfo ext4
name: ext4
filename: (builtin)
softdep: pre: crc32c
license: GPL
file: fs/ext4/ext4
description: Fourth Extended Filesystem
author: Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias: fs-ext4
alias: ext3
alias: fs-ext3
ksus@LAPTOP-BV8H0HF1:/tmp/test_dir$ |
```

2.8 Operating-System Structure

Struktur sistem operasi (**Operating-System Structure**) terdiri dari beberapa lapisan yang saling berhubungan, mulai dari **hardware**, **kernel**, **system calls**, **system programs**, hingga **user applications**. Lapisan hardware berisi perangkat fisik seperti CPU dan USB, yang bisa dieksplor dengan perintah `lspci`, `lsusb`, dan `cat /proc/cpuinfo`. Kernel berperan sebagai penghubung utama antara perangkat keras dan perangkat lunak, dapat dilihat dengan `uname -a`. Lapisan **system calls** seperti `strace` `ls` memungkinkan aplikasi berkomunikasi dengan kernel. Lapisan **system programs** menyediakan alat bantu seperti `ps` `aux` dan `top`, sedangkan **user applications** seperti `bash` dan `gcc` berada di lapisan tertinggi.

Selain itu, praktik membandingkan **monolithic kernel** yang memuat semua layanan inti dalam satu sistem besar dengan **microkernel** yang lebih modular dan efisien. Program `procfs_read.c` digunakan untuk membaca informasi CPU, memori, dan proses melalui direktori virtual `/proc`, menunjukkan bagaimana Linux mengorganisasi dan menyediakan akses terstruktur terhadap komponen sistem operasi.

2.8.1 Eksplorasi Struktur Layering

- **Layer 1: Hardware**

```

asus@LAPTOP-BV0A0HFI:~$ cat /tmp/test_dir$ lspci
1782:00:00:0 3D controller: Microsoft Corporation Basic Render Driver
db8e:00:00:0 SCSI storage controller: Red Hat, Inc. Virtio file system (rev 01)
asus@LAPTOP-BV0A0HFI:~$ lspci -vv
asus@LAPTOP-BV0A0HFI:~$ cat /proc/cpuinfo
processor       : 0
vendor_id      : AuthenticAMD
cpu family     : 68
model          : AMD Ryzen 7 7435HS
stepping       : 1
microcode     : 0xffffffff
cpu MHz        : 3904.099
cache size     : 512 KB
physical id    : 0
siblings       : 16
core id        : 0
cpu cores      : 8
apicid         : 0
initial apicid : 0
fpu            : yes
fpu_exception  : yes
cpuid level    : 13
wp             : yes
flags           : fpu vme de pse tsc mtr pae mce cx8 apic sep mtrr pge mca cmov pat pse3d clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpeibgt rdtscln constant tsc rep_good nopl tsc_reliable nonstop_tsc cpuid mwait_16x pclmulqdq ssse3 fma cpi sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm crat_legacy abm sse4a misalignsse 3dnowprefetch osvw
vmmcall
topoext perfctr_core ssbd ibpb stlb vmcall fsgsbase bti avx2 smep bmi2 erms invpcid rdtscp avx512f avx512vbmi avx512vbmi2 avx512vp avx512vqbmi avx512vqbi avx512vqbmi2 avx512vqbmi3 avx512vqbmi4 avx512vqbmi5 avx512vqbmi6 avx512vqbmi7 avx512vqbmi8 avx512vqbmi9 avx512vqbmi10 avx512vqbmi11 avx512vqbmi12 avx512vqbmi13 avx512vqbmi14 avx512vqbmi15 avx512vqbmi16 avx512vqbmi17 avx512vqbmi18 avx512vqbmi19 avx512vqbmi20 avx512vqbmi21 avx512vqbmi22 avx512vqbmi23 avx512vqbmi24 avx512vqbmi25 avx512vqbmi26 avx512vqbmi27 avx512vqbmi28 avx512vqbmi29 avx512vqbmi30 avx512vqbmi31 avx512vqbmi32 avx512vqbmi33 avx512vqbmi34 avx512vqbmi35 avx512vqbmi36 avx512vqbmi37 avx512vqbmi38 avx512vqbmi39 avx512vqbmi40 avx512vqbmi41 avx512vqbmi42 avx512vqbmi43 avx512vqbmi44 avx512vqbmi45 avx512vqbmi46 avx512vqbmi47 avx512vqbmi48 avx512vqbmi49 avx512vqbmi50 avx512vqbmi51 avx512vqbmi52 avx512vqbmi53 avx512vqbmi54 avx512vqbmi55 avx512vqbmi56 avx512vqbmi57 avx512vqbmi58 avx512vqbmi59 avx512vqbmi60 avx512vqbmi61 avx512vqbmi62 avx512vqbmi63 avx512vqbmi64 avx512vqbmi65 avx512vqbmi66 avx512vqbmi67 avx512vqbmi68 avx512vqbmi69 avx512vqbmi70 avx512vqbmi71 avx512vqbmi72 avx512vqbmi73 avx512vqbmi74 avx512vqbmi75 avx512vqbmi76 avx512vqbmi77 avx512vqbmi78 avx512vqbmi79 avx512vqbmi80 avx512vqbmi81 avx512vqbmi82 avx512vqbmi83 avx512vqbmi84 avx512vqbmi85 avx512vqbmi86 avx512vqbmi87 avx512vqbmi88 avx512vqbmi89 avx512vqbmi90 avx512vqbmi91 avx512vqbmi92 avx512vqbmi93 avx512vqbmi94 avx512vqbmi95 avx512vqbmi96 avx512vqbmi97 avx512vqbmi98 avx512vqbmi99 avx512vqbmi100 avx512vqbmi101 avx512vqbmi102 avx512vqbmi103 avx512vqbmi104 avx512vqbmi105 avx512vqbmi106 avx512vqbmi107 avx512vqbmi108 avx512vqbmi109 avx512vqbmi110 avx512vqbmi111 avx512vqbmi112 avx512vqbmi113 avx512vqbmi114 avx512vqbmi115 avx512vqbmi116 avx512vqbmi117 avx512vqbmi118 avx512vqbmi119 avx512vqbmi120 avx512vqbmi121 avx512vqbmi122 avx512vqbmi123 avx512vqbmi124 avx512vqbmi125 avx512vqbmi126 avx512vqbmi127 avx512vqbmi128 avx512vqbmi129 avx512vqbmi130 avx512vqbmi131 avx512vqbmi132 avx512vqbmi133 avx512vqbmi134 avx512vqbmi135 avx512vqbmi136 avx512vqbmi137 avx512vqbmi138 avx512vqbmi139 avx512vqbmi140 avx512vqbmi141 avx512vqbmi142 avx512vqbmi143 avx512vqbmi144 avx512vqbmi145 avx512vqbmi146 avx512vqbmi147 avx512vqbmi148 avx512vqbmi149 avx512vqbmi150 avx512vqbmi151 avx512vqbmi152 avx512vqbmi153 avx512vqbmi154 avx512vqbmi155 avx512vqbmi156 avx512vqbmi157 avx512vqbmi158 avx512vqbmi159 avx512vqbmi160 avx512vqbmi161 avx512vqbmi162 avx512vqbmi163 avx512vqbmi164 avx512vqbmi165 avx512vqbmi166 avx512vqbmi167 avx512vqbmi168 avx512vqbmi169 avx512vqbmi170 avx512vqbmi171 avx512vqbmi172 avx512vqbmi173 avx512vqbmi174 avx512vqbmi175 avx512vqbmi176 avx512vqbmi177 avx512vqbmi178 avx512vqbmi179 avx512vqbmi180 avx512vqbmi181 avx512vqbmi182 avx512vqbmi183 avx512vqbmi184 avx512vqbmi185 avx512vqbmi186 avx512vqbmi187 avx512vqbmi188 avx512vqbmi189 avx512vqbmi190 avx512vqbmi191 avx512vqbmi192 avx512vqbmi193 avx512vqbmi194 avx512vqbmi195 avx512vqbmi196 avx512vqbmi197 avx512vqbmi198 avx512vqbmi199 avx512vqbmi200 avx512vqbmi201 avx512vqbmi202 avx512vqbmi203 avx512vqbmi204 avx512vqbmi205 avx512vqbmi206 avx512vqbmi207 avx512vqbmi208 avx512vqbmi209 avx512vqbmi210 avx512vqbmi211 avx512vqbmi212 avx512vqbmi213 avx512vqbmi214 avx512vqbmi215 avx512vqbmi216 avx512vqbmi217 avx512vqbmi218 avx512vqbmi219 avx512vqbmi220 avx512vqbmi221 avx512vqbmi222 avx512vqbmi223 avx512vqbmi224 avx512vqbmi225 avx512vqbmi226 avx512vqbmi227 avx512vqbmi228 avx512vqbmi229 avx512vqbmi230 avx512vqbmi231 avx512vqbmi232 avx512vqbmi233 avx512vqbmi234 avx512vqbmi235 avx512vqbmi236 avx512vqbmi237 avx512vqbmi238 avx512vqbmi239 avx512vqbmi240 avx512vqbmi241 avx512vqbmi242 avx512vqbmi243 avx512vqbmi244 avx512vqbmi245 avx512vqbmi246 avx512vqbmi247 avx512vqbmi248 avx512vqbmi249 avx512vqbmi250 avx512vqbmi251 avx512vqbmi252 avx512vqbmi253 avx512vqbmi254 avx512vqbmi255 avx512vqbmi256 avx512vqbmi257 avx512vqbmi258 avx512vqbmi259 avx512vqbmi260 avx512vqbmi261 avx512vqbmi262 avx512vqbmi263 avx512vqbmi264 avx512vqbmi265 avx512vqbmi266 avx512vqbmi267 avx512vqbmi268 avx512vqbmi269 avx512vqbmi270 avx512vqbmi271 avx512vqbmi272 avx512vqbmi273 avx512vqbmi274 avx512vqbmi275 avx512vqbmi276 avx512vqbmi277 avx512vqbmi278 avx512vqbmi279 avx512vqbmi280 avx512vqbmi281 avx512vqbmi282 avx512vqbmi283 avx512vqbmi284 avx512vqbmi285 avx512vqbmi286 avx512vqbmi287 avx512vqbmi288 avx512vqbmi289 avx512vqbmi290 avx512vqbmi291 avx512vqbmi292 avx512vqbmi293 avx512vqbmi294 avx512vqbmi295 avx512vqbmi296 avx512vqbmi297 avx512vqbmi298 avx512vqbmi299 avx512vqbmi300 avx512vqbmi301 avx512vqbmi302 avx512vqbmi303 avx512vqbmi304 avx512vqbmi305 avx512vqbmi306 avx512vqbmi307 avx512vqbmi308 avx512vqbmi309 avx512vqbmi310 avx512vqbmi311 avx512vqbmi312 avx512vqbmi313 avx512vqbmi314 avx512vqbmi315 avx512vqbmi316 avx512vqbmi317 avx512vqbmi318 avx512vqbmi319 avx512vqbmi320 avx512vqbmi321 avx512vqbmi322 avx512vqbmi323 avx512vqbmi324 avx512vqbmi325 avx512vqbmi326 avx512vqbmi327 avx512vqbmi328 avx512vqbmi329 avx512vqbmi330 avx512vqbmi331 avx512vqbmi332 avx512vqbmi333 avx512vqbmi334 avx512vqbmi335 avx512vqbmi336 avx512vqbmi337 avx512vqbmi
```

- **Layer 2: Kernel**

```
asus@LAPTOP-BV84QHF1:/tmp/test_dir$ uname -a
Linux LAPTOP-BV84QHF1 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:38:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
asus@LAPTOP-BV84QHF1:/tmp/test_dir$ cat /proc/sys/kernel/ostype
Linux
asus@LAPTOP-BV84QHF1:/tmp/test_dir$
```

- **Layer 3: System Calls**

[illegible]

- **Layer 4: System Programs**

```
asus@LAPTOP-BV040HF1: /tm
asus@LAPTOP-BV040HF1:/tmp/test_dir$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1 22888 12236 ?        Ss   Oct13 0:02 /sbin/init
root         2  0.0  0.0   3872 1664 ?        Sl   Oct13 0:00 /init
root         7  0.0  0.0   3872 1792 ?        Sl   Oct13 0:00 /usr/sbin/sshd -o "PermitRootLogin=no"
root        44  0.0  0.2 66816 17148 ?        Ss+  Oct13 0:00 /usr/lib/systemd/systemd-journald
root        91  0.0  0.0 24876 6816 ?        Ss   Oct13 0:01 /usr/lib/systemd/systemd-devd
systemd+  111  0.0  0.1 21456 12672 ?        Ss   Oct13 0:00 /usr/lib/systemd/systemd-resolved
systemd+  134  0.0  0.0 91824 7552 ?        Ssl  Oct13 0:00 /usr/lib/systemd/systemd-timesyncd
root       160  0.0  0.0  4236 2568 ?        Ss   Oct13 0:00 /usr/sbin/cron -f -s
message+  161  0.0  0.0  9596 4864 ?        Ss   Oct13 0:00 dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root       168  0.0  0.1 17964 8128 ?        Ss   Oct13 0:00 /usr/lib/systemd/systemd-logind
root       170  0.0  0.1 175696 12888 ?        Ssl  Oct13 0:00 /usr/libexec/ssl-pro-service -vv
root       172  0.0  0.0   2168 1728 hvcc0 Ss+  Oct13 0:00 /sbin/agetty --p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
syslog    178  0.0  0.0 222588 5128 ?        Ssl  Oct13 0:00 /usr/sbin/rsyslogd -n -iNONE
root      192  0.0  0.0   3116 1792 tty1    Ss+  Oct13 0:00 /sbin/agetty --p -- \u --noclear - linux
root      193  0.0  0.2 187832 22772 ?        Ssl  Oct13 0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root      282  0.0  0.0   3876 896 ?         Ss   Oct13 0:00 /init
root      283  0.0  0.0   3892 1152 ?        S   Oct13 0:00 /init
asus      284  0.0  0.0  6872 5128 pts/0    Ss   Oct13 0:00 -bash
root      285  0.0  0.0   6816 4352 pts/1    Ss   Oct13 0:00 /bin/login -f
asus     328  0.0  0.1 28132 11904 ?        Ss   Oct13 0:00 /usr/lib/systemd/systemd --user
asus     329  0.0  0.0 21152 3528 ?         S   Oct13 0:00 (sd-pam)
asus     346  0.0  0.0   6872 5128 pts/1    Ss   Oct13 0:00 -bash
asus     408  0.0  0.0   2688 1408 pts/0    T   Oct13 0:00 /simple_shell
root     665  0.0  0.0 14148 4784 pts/0    r   Oct13 0:00 sudo journalctl -u my_daemon
root     666  0.0  0.0 14148 2688 pts/2    Ss   Oct13 0:00 sudo journalctl -u my_daemon
root     667  0.0  0.0 17876 7784 pts/2    T+  Oct13 0:00 journalctl -u my_daemon
root     668  0.0  0.0   3312 2384 pts/2    Ss   Oct13 0:00 less
root     909  0.0  0.0   3876 1828 ?        Ss   Oct13 0:00 /init
root     911  0.0  0.0   3892 1168 ?        S   Oct13 0:00 /init
asus     916  0.0  0.0  6872 3248 pts/3    Ss   Oct13 0:00 -bash
root     958  0.0  0.0 14152 4784 pts/3    Ss+ 00:00 0:00 sudo nano Makefile
root     959  0.0  0.0 14152 2876 pts/4    Ss+ 00:00 0:00 sudo nano Makefile
root     952  0.0  0.0  4688 3712 pts/4    Ss+ 00:00 0:00 nano Makefile
root    1289  0.0  0.2 378888 26352 ?        Ssl 00:15 0:00 /usr/libexec/packagelint
polkitd  1285  0.0  0.0 308164 7888 ?        Ssl 00:15 0:00 /usr/lib/polkit-1/polkitd --no-debug
asus    1427  0.0  0.0   8288 4896 pts/0    R+  00:18 0:00 ps aux

asus@LAPTOP-BV040HF1:/tmp/test_dir$ top
top - 00:18:39 up 3:20, 1 user, load average: 0.00, 0.00, 0.00
task: 36 total, 1 running, 22 sleeping, 3 stopped, 0 zombie
%cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7848.2 total, 7224.6 free, 561.5 used, 217.7 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7280.0 avail Mem

   PID USER      PR NI  VIRT  RES  SHR S  %CPU  %MEM    TIME+ COMMAND
    1 root        20  0 22888 12236 9292 S   0.0   0.2   0:02.07 systemd
    2 root        20  0  3872  1664 1664 S   0.0   0.0   0:00.01 init-systemd(
    7 root        20  0  3872  1792 1792 S   0.0   0.0   0:00.00 /init
   44 root       19 -1 66816 17148 16244 S   0.0   0.2   0:00.04 systemd-journ
    91 root       20  0 24876  6816 4864 S   0.0   0.1   0:01.25 systemd-devd
  111 systemd+  20  0 21456 12672 10496 S   0.0   0.2   0:00.15 systemd-resolve
  134 systemd+  20  0 21456 12672 10496 S   0.0   0.2   0:00.27 systemd-timesyn
  160 root       20  0  4236  2568 2432 S   0.0   0.0   0:00.03 /cron
  161 message+  20  0  9596  4864 4352 S   0.0   0.1   0:00.00 dbus-daemon
  168 root       20  0 17964  8128 7824 S   0.0   0.1   0:00.21 systemd-logind
  170 root       20  0 175696 12888 16248 S   0.0   0.2   0:00.39 ssl-pro-service
  172 root       20  0  2168  1728 1792 S   0.0   0.0   0:00.01 agetty
  178 syslog    20  0 222588 5128 4488 S   0.0   0.1   0:00.20 rsyslogd
  192 root       20  0  3116  1792 1664 S   0.0   0.0   0:00.00 agetty
  193 root       20  0 187832 22772 11184 S   0.0   0.3   0:00.00 unattended-upgr
  282 root       20  0  3876  896 896 S   0.0   0.0   0:00.00 SessionLeader
  283 root       20  0  3892 1152 824 S   0.0   0.0   0:00.03 /init(283)
 284 asus        20  0  6872 5128 3456 S   0.0   0.1   0:00.23 bash
```

- **Layer 5: User Applications**

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ which bash
/usr/bin/bash
asus@LAPTOP-BV040HF1:/tmp/test_dir$ which gcc
/usr/bin/gcc
asus@LAPTOP-BV040HF1:/tmp/test_dir$
```

2.8.2 Microkernel vs Monolithic

- **Lihat ukuran kernel (Monolithic)**

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ ls -lh /boot/vmlinuz-$(uname -r)
ls: cannot access '/boot/vmlinuz-6.6.87.2-microsoft-standard-WSL2': No such file or directory
```

- **Lihat built-in vs module**

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ cat /boot/config-$(uname -r) | grep "=y" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
asus@LAPTOP-BV040HF1:/tmp/test_dir$ cat /boot/config-$(uname -r) | grep "=m" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
```

- **asus@LAPTOP-BV040HF1:/tmp/test_dir\$**

- **Lihat loaded modules**

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ lsmod | wc -l
19
asus@LAPTOP-BV040HF1:/tmp/test_dir$
```

2.8.3 Program - Virtual File System

```
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ nano procfs_read.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ gcc -o procfs_read procfs_read.c
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ ./procfs_read

=== CPU Information ===
processor       : 0
vendor_id      : AuthenticAMD
cpu family     : 25
model          : 68
model name     : AMD Ryzen 7 7435HS
stepping       : 1
microcode      : 0xffffffff
cpu MHz        : 3094.099
cache size     : 512 KB
physical id    : 0

=== Memory Information ===
MemTotal:      8036548 kB
MemFree:       7365416 kB
MemAvailable:  7456148 kB
Buffers:       3924 kB
Cached:        215044 kB

=== Process Information ===
Name:          procfs_read
Umask:         0022
State:         R (running)
Tgid:          1503
Ngid:          0
Pid:           1503
PPid:          284
TracerPid:     0
Uid:           1000 1000 1000 1000
Gid:           1000 1000 1000 1000
asus@LAPTOP-BV04QHF1:/tmp/test_dir$ |
```

2.9 Building and Booting an Operating System

Pada praktik Building and Booting an Operating System, peserta mempelajari proses penting dalam menjalankan sistem Linux sejak komputer dinyalakan hingga sistem siap digunakan. Tahapan dimulai dari BIOS/UEFI yang mengeksekusi bootloader GRUB, yang bertugas memuat kernel Linux dan initramfs ke memori. Kernel kemudian menginisialisasi perangkat keras dan memulai proses utama sistem seperti systemd. Perintah dmesg digunakan untuk melihat pesan kernel selama booting, sementara systemd-analyze membantu menganalisis waktu dan urutan layanan saat startup. File /boot/grub/grub.cfg berisi konfigurasi GRUB yang menentukan opsi boot. Selain itu, parameter kernel yang aktif dapat dilihat dengan cat /proc/cmdline, dan dapat diubah sementara melalui menu GRUB atau secara permanen dengan mengedit file /etc/default/grub lalu menjalankan sudo update-grub. Praktik ini membantu memahami struktur dasar dan mekanisme kerja sistem operasi dari awal proses boot hingga sistem siap digunakan.

2.9.1 Analisis Boot Process

- **Lihat boot messages**

```

[ 0.000000] Linux version 6.6.82-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_
DYNAMIC Thu Jun 5 18:30:46 UTC 2025
[ 0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_c
ount=0 WSL_ENABLE_CRASH_DUMP=1
[ 0.000000] KERNEL supported cpus:
[ 0.000000] Intel GenuineIntel
[ 0.000000] AMD AuthenticAMD
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000e0000-0x00000000000e0fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000200000-0x00000000007fffff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000010000000-0x000000002023ffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] DMI not present or invalid.
[ 0.000000] Hypervisor detected: Microsoft Hyper-V
[ 0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x900c2c, misc 0xe0bed7b6
[ 0.000000] Hyper-V: Nested features: 0x4a0000
[ 0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[ 0.000000] Hyper-V: Using hypercall for remote TLB flush
[ 0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] tsc: Detected 3094.099 MHz processor
[ 0.00069] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[ 0.00072] e820: remove [mem 0x000a0000-0x000fffff] usable
[ 0.00075] last_pfn = 0x202400 max_arch_pfn = 0x400000000
[ 0.00099] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[ 0.00102] x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
[ 0.00142] last_pfn = 0xf8000 max_arch_pfn = 0x400000000
[ 0.00154] Using GB pages for direct mapping
[ 0.00278] RAMDISK: [mem 0x04be4000-0x04effff]
[ 0.00281] ACPI: Early table checksum verification disabled
[ 0.00284] ACPI: RSDP 0x000000000000E000 000024 (v02 VIRTUAL)
[ 0.00288] ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 NSFT 00000001)
[ 0.00293] ACPI: FACP 0x0000000000101000 000114 (v06 VIRTUAL MICROSOFT 00000001 NSFT 00000001)
[ 0.00296] ACPI: DSDT 0x00000000001011B8 01E11C (v02 MSFTVM DSDT01 00000001 INTL 20230628)

```

- **Lihat systemd boot analysis**

```

asus@LAPTOP-BV04OHHF:~$
asus@LAPTOP-BV04OHHF:~$ mp/test_dir1 systemd-analyze
Startup finished in 1.074s (userspace)
graphical.target reached after 1.436s in userspace.
asus@LAPTOP-BV04OHHF:~$ mp/test_dir1 systemd-analyze blame
42ms systemd.service
77ms landscope-client.service
11ms snapd.socket.service
11ms dev-sdd.device
20ms snapd.service
21ms systemd-udev-trigger.service
21ms systemd-udevd.service
18ms user@1000.service
18ms wsl-pro.service
11ms systemd-resolved.service
6ms rsyslog.service
72ms systemd-tapfiles-clean.service
8ms logrotate.service
60ms systemd-timesyncd.service
67ms systemd-journal-flush.service
89ms keyboard-setup.service
58ms systemd-journal.service
97ms systemd-logind.service
55ms systemd-udevd.service
33ms polkit.service
46ms dpkg-db-backup.service
49ms cifs.mount.service
42ms systemd-tapfiles-setup.service
13ms user-runtime-dir@1000.service
13ms dbus.service
33ms dev-hugepages.mount
32ms dev-mqueue.mount
11ms snapd.socket
21ms sys-kernel-tracing.mount
58ms sys-kernel-tracing.mount
26ms systemd-tapfiles-setup-dev-early.service
26ms systemd-binfmt.service
20ms kmod-static-nodes.service
23ms mdprobe@conf.ifrs.service
23ms mdprobe@kds_md.service
23ms mdprobe@kds.service
23ms mdprobe@el_ptyone.service
[time 1.27]
asus@LAPTOP-BV04OHHF:~$ mp/test_dir1 systemd-analyze blame
asus@LAPTOP-BV04OHHF:~$ mp/test_dir1 systemd-analyze critical-chain
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "." character.
graphical.target @1.436s
├─multi-user.target @1.436s
│ └─snapd.socket @60ms
│   └─basic.target @61ms
│     └─sockets.target @61ms
│       └─snapd.socket @61ms
│         └─systemd.target @61ms
│           └─systemd-resolved.service @60ms
│             └─systemd-tapfiles-setup.service @61ms
│               └─systemd-journal-flush.service @7ms
│                 └─systemd-journal.service @26ms
│                   └─system-journal.socket @22ms
└─system.slice @22ms
    └─ttt @22ms
asus@LAPTOP-BV04OHHF:~$ mp/test_dir1

```

- **Lihat GRUB configuration**

```
asus@LAPTOP-BV040HF1:/tmp/test_dir$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory
asus@LAPTOP-BV040HF1:/tmp/test_dir$
```

- **Lihat initramfs**

```

asus@LAPTOP-BV040HF1:/tmp/test_dir$ lsinitramfs /boot/initrd.img-$uname -r | head -20
/usr/bin/unkninitramfs: 64: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
/usr/bin/unkninitramfs: 57: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
/usr/bin/unkninitramfs: 38: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
cpio: premature end of archive
asus@LAPTOP-BV040HF1:/tmp/test_dir$

```

2.9.2 Using strace

- **Trace system calls**

[illegible]

- **Trace dengan detail**

[illegible]

- **Hitung waktu system calls**

```

$ sudo strace -c ls
Makefile      hello_module.c  libmathops.so  main.o        math_ops.h     math_ops.s     os_specific.c  procfs_read    program_dynamic  simple_shell
file_syscalls info_syscalls   main.c         main.s        math_ops.i     my_daemon      process_syscalls  procfs_read.c  program_shared  simple_shell.c
file_syscalls.c info_syscalls.c main.i         math_ops.c    math_ops.o     my_daemon.c    process_syscalls.c  program        program_static  simple_shell.c
% time        seconds         usecs/call    calls         errors         syscall
-----
26.56 0.002021      57           35           13 openat
20.93 0.001593      53           30           mmap
12.29 0.000935      38           24           close
19.74 0.000817      35           23           fstat
7.54 0.000574     57           10           read
5.48 0.000417     208          2           pread64
3.60 0.000274      91           3           write
3.56 0.000271      54           5           mprotect
1.31 0.000100      100          1           arch_prctl
1.31 0.000100      50           2           getdents64
1.25 0.000095     47           2           1 statfs
0.97 0.000074      37           2           1 ioctl
0.74 0.000056      56           1           munmap
0.60 0.000046      15           3           brk
0.60 0.000046      23           2           2 access
0.49 0.000037      37           1           futex
0.42 0.000032      32           1           prlimit64
0.42 0.000032      32           1           getrandom
0.39 0.000030      30           1           set_tid_address
0.39 0.000030      30           1           set_robust_list
0.39 0.000030      30           1           rseq
0.00 0.000000      0            1           execve
-----
100.00 0.007610      50           152          16 total

```


- **Trace dan simpan ke file**

[illegible]

2.9.3 Using ltrace

- **Trace library calls**

```

source($ENV{HOME}/bin/perl:/tmp/test_dir) { trace is
    strchr("ls", "/")
        = nil
    setlocale(LC_ALL, "")
        = "LC.UTF-8"
    bindtextdomain("coreutils", "/usr/share/locale")
        = "/usr/share/locale"
    textdomain("coreutils")
        = "coreutils"
    _cxa_atexit(0x5fcd90880, 0x5fcd902008, 0)
        = 0
    getopt_long(1, 0x7ffdc079108, "abcdgfhlmnopqrstuvw:XABCDGFIH:... ", 0x5fcd901f300, -1)
        = -1
    getenv("LS_BLOCK_SIZE")
        = nil
    getenv("_BLOCK_SIZE")
        = nil
    getenv("_BLOCKSIZE")
        = nil
    getenv("POSIXLY_CORRECT")
        = nil
    getenv("_BLOCK_SIZE")
        = nil
    isatty(1)
        = 1
    ioctl(1, 23023, 0x7ffdc0478f20)
        = 0
    getenv("_TABSIZE")
        = nil
    getenv("QUOTING_STYLE")
        = nil
    _errno_location()
        = 0x7b59d64776a0
    _errno_location()
        = 0x7b59d64776a0
    getenv("TZ")
        = nil
    reallocarray(0, 100, 200, 0x5fd1qc406c0)
        = 0x5fd1qc406d0
    strlen("")
        = 1
    __memcpy_chk(0x5fd1qc45850, 0x5fcd901fdec, 2, 2)
        = 0x5fd1qc45850
    _errno_location()
        = 0x7b59d64776a0
    opendir("/")
        = 0x5fd1qc45970
    readdir(0x5fd1qc45870)
        = 0x5fd1qc458a0
    _errno_location()
        = 0x7b59d64776a0
    _ctype_get_mb_cur_max()
        = 6
    strlen("file.syscalls.c")
        = 15
    strlen("file.syscalls.c")
        = 15
    __memcpy_chk(0x5fd1qc4db00, 0x5fd1qc458b3, 16, 16)
        = 0x5fd1qc4db00
    readdir(0x5fd1qc45870)
        = 0x5fd1qc458c0
    _errno_location()
        = 0x7b59d64776a0
    _ctype_get_mb_cur_max()
        = 6
    strlen("math_ops.i")
        = 10
    strlen("math_ops.i")
        = 10
    __memcpy_chk(0x5fd1qc4db00, 0x5fd1qc458db, 11, 11)
        = 0x5fd1qc4db00
    readdir(0x5fd1qc45870)
        = 0x5fd1qc458e0
    _errno_location()
        = 0x7b59d64776a0
    _ctype_get_mb_cur_max()
        = 6
    strlen("process.syscalls.c")
        = 18
    strlen("process.syscalls.c")
        = 18
    __memcpy_chk(0x5fd1qc4db00, 0x5fd1qc458fb, 19, 19)
        = 0x5fd1qc4db00
    readdir(0x5fd1qc45870)
        = 0x5fd1qc45910
    _errno_location()
        = 0x7b59d64776a0
    _ctype_get_mb_cur_max()
        = 6
    strlen("math_ops.c")
        = 10
    strlen("math_ops.c")
        = 10
    __memcpy_chk(0x5fd1qc4d910, 0x5fd1qc45923, 11, 11)
        = 0x5fd1qc4d910
    readdir(0x5fd1qc45870)
        = 0x5fd1qc45930
    _errno_location()
        = 0x7b59d64776a0
    _ctype_get_mb_cur_max()
        = 6
    strlen("program.static")
        = 14
    strlen("program.static")
        = 14
    __memcpy_chk(0x5fd1qc4d930, 0x5fd1qc45943, 15, 15)
        = 0x5fd1qc4d930

```


- Dengan detail

```
asus@LAPTOP-BVB4QHF1:/tmp/test_dir$ ltrace -c ls
ls
-----
% time seconds usecs/call calls function
-----
19.34 0.040225 325 151 _errno_location
15.79 0.040191 321 125 strlen
14.07 0.037838 323 117 strcoll
12.58 0.031813 342 93 _ctype_get_mb_cur_max
11.88 0.028209 357 79 __overflow
6.92 0.017613 17613 1 setlocale
4.58 0.011402 316 34 readdir
4.17 0.010693 342 31 fwrite_unlocked
4.01 0.010201 318 32 __memcpy_chk
1.93 0.000917 327 15 memcpy
0.89 0.002276 284 8 getenv
0.56 0.001122 355 4 reallocarray
0.48 0.001227 366 4 _freading
0.47 0.001189 594 2 fclose
0.32 0.000827 413 2 __fpending
0.28 0.000705 765 1 opendir
0.25 0.000633 316 2 fflush
0.25 0.000631 631 1 ioctl
0.24 0.000602 361 2 fileno
0.19 0.000495 495 1 strrchr
0.19 0.000495 495 1 closedir
0.18 0.000457 457 1 isatty
0.12 0.000317 317 1 _cxa_atexit
0.12 0.000314 314 1 bindtextdomain
0.11 0.000286 286 1 setjmp
0.11 0.000283 283 1 getopt_long
0.11 0.000273 273 1 textdomain
-----
100.00 0.254484 712 total
asus@LAPTOP-BVB4QHF1:/tmp/test_dir$
```

2.9.4 Using GDB

- Debug dengan GDB

```
asus@LAPTOP-BVB4QHF1:/tmp/test_dir$ nano buggy.c
asus@LAPTOP-BVB4QHF1:/tmp/test_dir$
```

- Compile dengan debug symbols

```
asus@LAPTOP-BVB4QHF1:/tmp/test_dir$ gcc -g -o buggy buggy.c
asus@LAPTOP-BVB4QHF1:/tmp/test_dir$
```

- Di dalam GDB

```
asus@LAPTOP-BVB4QHF1:/tmp/test_dir$ gdb ./buggy
GNU gdb (Ubuntu 15.0.50-2624003-ubuntu1) 15.0.50.2624003-gi
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./buggy...
(gdb) break main
Breakpoint 1 at 0x1238: file buggy.c, line 19.
(gdb) run
Starting program: /tmp/test_dir/buggy
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at buggy.c:19
19      buggy_function(5);
(gdb) next
array[0] = 0
array[1] = 2
array[2] = 0
array[3] = 0
array[4] = 8
20      return 0;
(gdb) step
21
(gdb) print n
No symbol "n" in current context.
(gdb) continue
Continuing.
[Inferior 1 (process 2928) exited normally]
(gdb) backtrace
No stack.
(gdb) quit
```

2.9.5 Kernel Debugging dengan dmesg

- Lihat kernel log

```
asus@LAPTOP-BV04OHP1:~$ dmesg
[0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
[0.000000] Command line: initrd=initrd.img WSL_ROOT_INIT=1 panic=1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty_legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
[0.000000] KERNEL supported cpus:
[0.000000] Intel GenuineIntel
[0.000000] AMD AuthenticAMD
[0.000000] BIOS-provided physical RAM map:
[0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] usable
[0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] reserved
[0.000000] BIOS-e820: [mem 0x0000000001000000-0x000000000000ffff] ACPI data
[0.000000] BIOS-e820: [mem 0x0000000002000000-0x000000000000ffff] usable
[0.000000] BIOS-e820: [mem 0x0000000100000000-0x000000002023ffff] usable
[0.000000] NX (Execute Disable) protection: active
[0.000000] APIC: Static calls initialized
[0.000000] DMI not present or invalid.
[0.000000] Hypervisor detected: Microsoft Hyper-V
[0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b0830, hints 0x900c2c, misc 0xe0bed7b6
[0.000000] Hyper-V: Nested features: 0x4a0800
[0.000000] Hyper-V: LAPIC timer frequency: 0xc3500
[0.000000] Hyper-V: Using hypercall for remote TLB flush
[0.000000] clocksource: hyperv_clocksource.tsc_page: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 400795202120 ns
[0.000000] clocksource: hyperv_clocksource.msrt: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 400795202120 ns
[0.000000] tsc: Detected 3094.099 MHz processor
[0.000000] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[0.000000] e820: remove [mem 0x00000000-0x00000fff] usable
[0.000000] last_pfn = 0x2024000 max_arch_pfn = 0x400000000
[0.000000] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[0.000000] x86/PAT: Configuration [0-7]: WB WC UC-UC WB WP UC-WT
[0.000000] last_pfn = 0xf0000 max_arch_pfn = 0x400000000
[0.000000] Using GB pages for direct mapping
[0.000000] RAMDISK: [mem 0x00000000-0x00000fff]
[0.000000] ACPI: Early table checksum verification disabled
[0.000000] ACPI: RSDP 0x0000000000000000 000024 (v02 VIRTUAL)
[0.000000] ACPI: XSDT 0x0000000000000000 000004 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[0.000000] ACPI: FACP 0x0000000000010000 000011 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[0.000000] ACPI: DSDT 0x0000000000010110 01E11C (v02 MSFTVM DSDT01 00000001 INTL 20230628)
[0.000000] ACPI: FACS 0x0000000000010114 000000
[0.000000] ACPI: OEM0 0x0000000000010154 000004 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[0.000000] ACPI: SRAT 0x000000000001F2D4 000000 (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[0.000000] ACPI: APIC 0x000000000001F684 0000C8 (v04 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[0.000000] ACPI: Reserving FACP table memory at [mem 0x101000-0x101113]
[0.000000] ACPI: Reserving DSDT table memory at [mem 0x1011B8-0x11F2D3]
[0.000000] ACPI: Reserving FACS table memory at [mem 0x101114-0x101153]
[0.000000] ACPI: Reserving OEM0 table memory at [mem 0x101154-0x1011D7]
[0.000000] ACPI: Reserving SRAT table memory at [mem 0x11F2D4-0x11F683]
[0.000000] ACPI: Reserving APIC table memory at [mem 0x11F684-0x11F74B]
[0.000000] SRAT: PXM 0 -> APIC 0x00 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x01 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x02 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x03 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x04 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x05 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x06 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x07 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x08 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x09 -> Node 0
```

- Follow kernel log

```
asus@LAPTOP-BV04OHP1:~$ dmesg -t
[0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
[0.000000] Command line: initrd=initrd.img WSL_ROOT_INIT=1 panic=1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty_legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
[0.000000] KERNEL supported cpus:
[0.000000] Intel GenuineIntel
[0.000000] AMD AuthenticAMD
[0.000000] BIOS-provided physical RAM map:
[0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] usable
[0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] reserved
[0.000000] BIOS-e820: [mem 0x0000000001000000-0x000000000000ffff] ACPI data
[0.000000] BIOS-e820: [mem 0x0000000002000000-0x000000000000ffff] usable
[0.000000] BIOS-e820: [mem 0x0000000100000000-0x000000002023ffff] usable
[0.000000] NX (Execute Disable) protection: active
[0.000000] APIC: Static calls initialized
[0.000000] DMI not present or invalid.
[0.000000] Hypervisor detected: Microsoft Hyper-V
[0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b0830, hints 0x900c2c, misc 0xe0bed7b6
[0.000000] Hyper-V: Nested features: 0x4a0800
[0.000000] Hyper-V: LAPIC timer frequency: 0xc3500
[0.000000] Hyper-V: Using hypercall for remote TLB flush
[0.000000] clocksource: hyperv_clocksource.tsc_page: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 400795202120 ns
[0.000000] clocksource: hyperv_clocksource.msrt: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 400795202120 ns
[0.000000] tsc: Detected 3094.099 MHz processor
[0.000000] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[0.000000] e820: remove [mem 0x00000000-0x00000fff] usable
[0.000000] last_pfn = 0x2024000 max_arch_pfn = 0x400000000
[0.000000] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[0.000000] x86/PAT: Configuration [0-7]: WB WC UC-UC WB WP UC-WT
[0.000000] last_pfn = 0xf0000 max_arch_pfn = 0x400000000
[0.000000] Using GB pages for direct mapping
[0.000000] RAMDISK: [mem 0x00000000-0x00000fff]
[0.000000] ACPI: Early table checksum verification disabled
[0.000000] ACPI: RSDP 0x0000000000000000 000024 (v02 VIRTUAL)
[0.000000] ACPI: XSDT 0x0000000000000000 000004 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[0.000000] ACPI: FACP 0x0000000000010000 000011 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[0.000000] ACPI: DSDT 0x0000000000010110 01E11C (v02 MSFTVM DSDT01 00000001 INTL 20230628)
[0.000000] ACPI: FACS 0x0000000000010114 000000
[0.000000] ACPI: OEM0 0x0000000000010154 000004 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[0.000000] ACPI: SRAT 0x000000000001F2D4 000000 (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[0.000000] ACPI: APIC 0x000000000001F684 0000C8 (v04 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[0.000000] ACPI: Reserving FACP table memory at [mem 0x101000-0x101113]
[0.000000] ACPI: Reserving DSDT table memory at [mem 0x1011B8-0x11F2D3]
[0.000000] ACPI: Reserving FACS table memory at [mem 0x101114-0x101153]
[0.000000] ACPI: Reserving OEM0 table memory at [mem 0x101154-0x1011D7]
[0.000000] ACPI: Reserving SRAT table memory at [mem 0x11F2D4-0x11F683]
[0.000000] ACPI: Reserving APIC table memory at [mem 0x11F684-0x11F74B]
[0.000000] SRAT: PXM 0 -> APIC 0x00 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x01 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x02 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x03 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x04 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x05 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x06 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x07 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x08 -> Node 0
[0.000000] SRAT: PXM 0 -> APIC 0x09 -> Node 0
```

- Filter by level

```

asus@LAPTOP-BV04OHP1:~$ dmesg -w
[10]+  Stopped                  dmesg -w
asus@LAPTOP-BV04OHP1:/tmp/test_dir$ dmesg -l err
0.111856] PCI: fatal: no config space access function found
1.522778] misc dxg: dxgk: dxgkio.is.feature.enabled: ioctl failed: -22
1.526923] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -22
1.527082] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -22
1.527873] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -22
1.528810] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
1.586682] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.203357] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.203961] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.204657] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.205203] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.205715] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.206348] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.206498] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.207044] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.207694] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.208457] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.208984] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
2.265697] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
5.798149] WSL (262) ERROR: CheckConnection: getaddrinfo() failed: -6
3109.731730] misc dxg: dxgk: dxgkio.is.feature.enabled: ioctl failed: -22
3109.732886] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -22
3109.737637] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -22
3109.738105] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -22
3109.738729] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3109.907523] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.142712] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.143341] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.148891] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.148841] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.148934] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.148843] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.148833] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.148809] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.148916] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.147556] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.148886] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
3150.258486] misc dxg: dxgk: dxgkio.query_adapter.info: ioctl failed: -2
6030.791796] WSL (262) ERROR: CheckConnection: getaddrinfo() failed: -6
6030.872362] WSL (262) ERROR: CheckConnection: getaddrinfo() failed: -6
asus@LAPTOP-BV04OHP1:/tmp/test_dir$ dmesg -l warn
0.098948] Speculative Return Stack Overflow: INPB-extending microcode not applied!
0.098948] Speculative Return Stack Overflow: WARNING: See https://kernel.org/doc/html/latest/admin-guide/hw-vuln/srso.html for mitigation options.
0.155285] PCI: System does not support PCI
0.208740] device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate IMA measurements will not be recorded in the IMA log.
1.525566] pulseaudio[1597]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
1.823272] systemd-journal[51]: File /var/log/journal/0d80b415eab0a3c92726d362daef8c/system.journal corrupted or uncleanly shut down, renaming and replacing.
3.925131] systemd-journal[51]: File /var/log/journal/0d80b415eab0a3c92726d362daef8c/user-1000.journal corrupted or uncleanly shut down, renaming and replacing.
2109.408190] TCP: nath: Driver has suspect GSO implementation, TCP performance may be compromised.
Exception:
3115.240658] Operation canceled @p9io.cpp:258 (AcceptAsync)
[ 3150.095588] systemd-journal[51]: File /var/log/journal/0d80b415eab0a3c92726d362daef8c/system.journal corrupted or uncleanly shut down, renaming and replacing.
asus@LAPTOP-BV04OHP1:/tmp/test_dir$

```

2.9.6 Memory Leak Detection dengan Valgrind

- Install valgrind

```

asus@LAPTOP-BV04OHP1:/tmp/test_dir$ sudo apt install valgrind
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm9
Use 'sudo apt autoremove' to remove it.
Suggested packages:
  valgrind-dbg valgrind-mpi kcachegrind alioyop valkyrie
The following NEW packages will be installed:
  valgrind
0 upgraded, 1 newly installed, 0 to remove and 13 not upgraded.
Need to get 14.9 MB of archives.
After this operation, 78.8 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 valgrind amd64 1:3.22.0-8ubuntu3 [14.9 MB]
Fetched 14.9 MB in 23s (662 kB/s)
Selecting previously unselected package valgrind.
(Reading database ... 58595 files and directories currently installed.)
Preparing to unpack .../valgrind_1:3.22.0-8ubuntu3_amd64.deb ...
Unpacking valgrind (1:3.22.0-8ubuntu3) ...
Setting up valgrind (1:3.22.0-8ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
asus@LAPTOP-BV04OHP1:/tmp/test_dir$

```

- Buat program dengan memory leak

```

asus@LAPTOP-BV04OHP1:/tmp/test_dir$ cat > memLeak.c << 'EOF'
> #include <stdlib.h>
int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forget to free(ptr)
    return 0;
}
EOF
asus@LAPTOP-BV04OHP1:/tmp/test_dir$ gcc -g -o memLeak memLeak.c
asus@LAPTOP-BV04OHP1:/tmp/test_dir$

```

- Jalankan dengan valgrind

```
asus@LAPTOP-BV04QHFI:/tmp/test_dir$ valgrind --leak-check=full ./memleak
==3683== Memcheck, a memory error detector
==3683== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==3683== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==3683== Command: ./memleak
==3683==
==3683== HEAP SUMMARY:
==3683==   in use at exit: 480 bytes in 1 blocks
==3683==   total heap usage: 1 allocs, 0 frees, 480 bytes allocated
==3683==
==3683== 480 bytes in 1 blocks are definitely lost in loss record 1 of 1
==3683==    at 0x484028: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==3683==    by 0x10915E: main (memleak.c:3)
==3683==
==3683== LEAK SUMMARY:
==3683==   definitely lost: 480 bytes in 1 blocks
==3683==   indirectly lost: 0 bytes in 0 blocks
==3683==   possibly lost: 0 bytes in 0 blocks
==3683==   still reachable: 0 bytes in 0 blocks
==3683==   suppressed: 0 bytes in 0 blocks
==3683==
==3683== For lists of detected and suppressed errors, rerun with: -s
==3683== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
asus@LAPTOP-BV04QHFI:/tmp/test_dir$
```

2.9.7 Performance Analysis

- CPU profiling dengan perf

```
asus@LAPTOP-BV04QHFI:/tmp/test_dir$ sudo apt install linux-tools-common linux-tools-generic
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  liblvm2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  hwdmdata libnumal libtraceevent1 linux-tools-6.8.0-85-generic linux-tools-6.8.0-85 linux-tools-6.8.0-85-generic linux-tools-common linux-tools-generic
The following NEW packages will be installed:
  hwdmdata libnumal libtraceevent1 linux-tools-6.8.0-85 linux-tools-6.8.0-85-generic linux-tools-common linux-tools-generic
0 upgraded, 7 newly installed, 0 to remove and 13 not upgraded.
Need to get 6303 kB of archives.
After this operation, 28.9 MB of additional disk space will be used.
Do you want to continue? [y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libnumal amd64 2.0.18-1build1 [23.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 hwdmdata all 0.379-1 [29.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-common all 6.8.0-85.85 [762 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-85 amd64 6.8.0-85.85 [5421 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-85-generic amd64 6.8.0-85.85 [1884 B]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-generic amd64 6.8.0-85.85 [19.0 kB]
Fetched 6303 kB in 18s (649 kB/s)
Selecting previously unselected package libnumal:amd64.
(Reading database ... 50967 files and directories currently installed.)
Preparing to unpack .../6-libnumal_2.0.18-1build1_amd64.deb ...
Unpacking libnumal:amd64 (2.0.18-1build1) ...
Selecting previously unselected package libtraceevent1:amd64.
Preparing to unpack .../7-libtraceevent1_1.10.1-2ubuntu2.1_amd64.deb ...
Unpacking libtraceevent1:amd64 (1:1.10.1-2ubuntu2.1) ...
Selecting previously unselected package hwdmdata.
Preparing to unpack .../2-hwdmdata_0.379-1_all.deb ...
Unpacking hwdmdata (0.379-1) ...
Selecting previously unselected package linux-tools-common.
Preparing to unpack .../3-linux-tools-common_6.8.0-85.85_all.deb ...
Unpacking linux-tools-common (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-6.8.0-85.
Preparing to unpack .../4-linux-tools-6.8.0-85_6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-6.8.0-85 (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-6.8.0-85-generic.
Preparing to unpack .../5-linux-tools-6.8.0-85-generic_6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-6.8.0-85-generic (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-generic.
Preparing to unpack .../6-linux-tools-generic_6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-generic (6.8.0-85.85) ...
Setting up hwdmdata (0.379-1) ...
Setting up libnumal:amd64 (2.0.18-1build1) ...
Setting up libtraceevent1:amd64 (1:1.10.1-2ubuntu2.1) ...
Setting up linux-tools-common (6.8.0-85.85) ...
Setting up linux-tools-6.8.0-85 (6.8.0-85.85) ...
Setting up linux-tools-6.8.0-85-generic (6.8.0-85.85) ...
Setting up linux-tools-generic (6.8.0-85.85) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu6) ...
```

- Monitor system calls

```
asus@LAPTOP-BV04QHFI:/tmp/test_dir$ sudo perf trace ls
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
asus@LAPTOP-BV04QHFI:/tmp/test_dir$
```

- Top-like interface

```
asus@LAPTOP-BV04QHFI:/tmp/test_dir$ sudo perf top
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
```

BAB III

PENUTUP

KESIMPULAN

Dari hasil praktikum Sistem Operasi yang telah dilakukan, dapat disimpulkan bahwa sistem operasi memiliki peran utama sebagai pengelola seluruh sumber daya komputer dan penghubung antara perangkat keras serta perangkat lunak. Melalui berbagai percobaan pada lingkungan Linux, mahasiswa dapat memahami cara kerja sistem operasi dalam mengatur proses, memori, file, serta komunikasi antarproses melalui layanan seperti *system calls* dan *system services*.

Selain itu, mahasiswa juga mampu mengenal struktur sistem operasi, proses *linking* dan *loading*, serta interaksi antara pengguna, shell, dan kernel. Praktikum ini memberikan pengalaman langsung dalam mengimplementasikan teori melalui pemrograman C sederhana, sehingga memperkuat pemahaman terhadap konsep manajemen sistem operasi. Secara keseluruhan, kegiatan ini berhasil meningkatkan wawasan dan keterampilan mahasiswa dalam memahami mekanisme internal sistem operasi modern serta penerapannya dalam dunia komputasi.